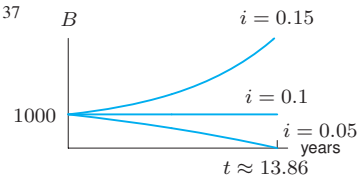
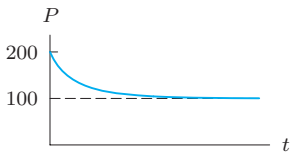
37



39 (a) $dB/dt = 0.05B - 12{,}000$
(b) $B = (B_0 - 240{,}000)e^{0.05t} + 240{,}000$
(c) \$151,708.93

41 $C(t) = 30e^{-100.33t}$

43 No, no



45 (a) $dx/dt = k(a-x)(b-x)$
(b) $x = ab(e^{bkt} - e^{akt})/(be^{bkt} - ae^{akt})$

47 (a) $x \to \infty$ exponentially
$y \to 0$ exponentially
(b) Predator-prey

49 (a) $x \to \infty$ exponentially
$y \to 0$ exponentially
(b) y is helped by the presence of x

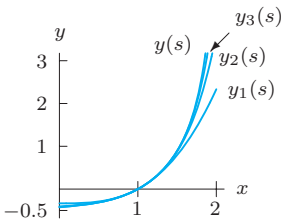
51 (a) $dy/dx = y(x+3)/(x(y+2))$
$y^2e^y = Ax^3e^x$
(d) $y^2e^y = x^3e^{x+4}$
(e) $y = x = e^{-4} \approx 0.0183$
(f) $y \approx e^{-13}$

53 (a) $x \to \infty$ if $y = 0$
$y \to 0$ if $x = 0$
(b) $(0, 0)$ and $(10{,}000, 150)$
(c) $dy/dx = y(-10 + 0.001x)/(x(3 - 0.02y))$
$3\ln y - 0.02y = -10\ln x + 0.001x + 94.1$
(f) $A \to B \to C \to D$
(h) At points A and C:
$dy/dx = 0$
At points B and D:
$dx/dy = 0$

55 P decreases to 0

57 P decreases towards $2L$

59 (c)

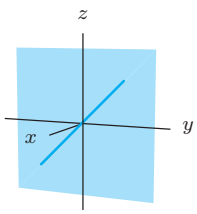


Section 12.1

1 Q

3 B, B, B

5



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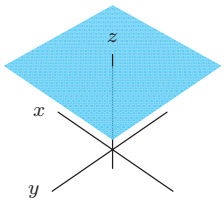

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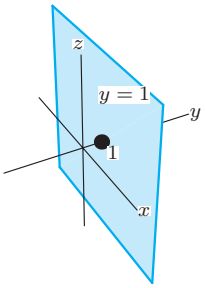
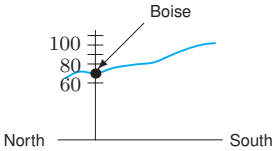


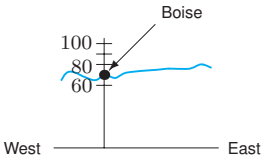
11 $(2, 3, 15)$

13 $(x-1)^2 + y^2 + z^2 = 4$

15 (a) 80-90°F
(b) 60-72°F
(c) 60-100°F

17





21 (a) −19°F
(b) 20 mph
(c) About 17.5 mph
(d) About 16.7°F

27 $(1.5,\ 0.5,\ -0.5)$

29 Cone, tip at origin, along x-axis with slope of 1

31 (a) $z = 7, z = -1$
(b) $x = 6, x = -2$
(c) $y = 7, y = -1$

33 (a) $(12, 7, 2); (5, 7, 2); (12, 1, 2)$
(b) $(5, 1, 4); (5, 7, 4); (12, 1, 4)$

35 (a) yz-plane: circle $(y+3)^2 + (z-2)^2 = 3$
xz-plane: none
xy-plane: point $(1, -3, 0)$
(b) Does not intersect

37 $(8, 0, \sqrt{3})$

39 $y = 1$ is a plane, not a line

41 Distance is 5

43 $(-2, -1, -5)$

45 True

47 False

49 True

51 True

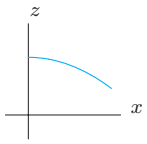
53 True

55 False

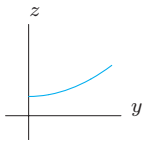
57 False

Section 12.2

1 (a) Decreases

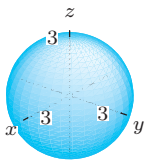


(b) Increases

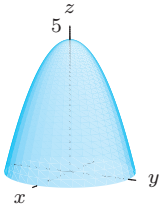


3 (a) I
(b) V
(c) IV
(d) II
(e) III

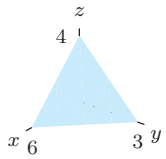
5 Sphere, radius 3



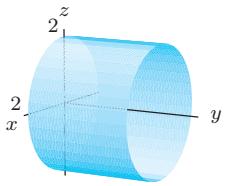
7 Upside-down bowl, vertex $(0, 0, 5)$



9 Plane, x-intercept 6, y-intercept 3, z-intercept 4

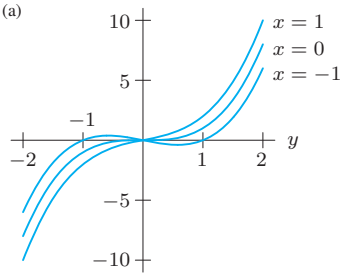


11 Circular cylinder extended in the y-direction

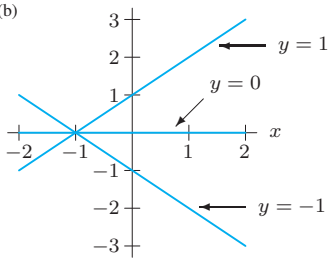


13 $x^2 + (y - \sqrt{7})^2 + z^2 = 9$

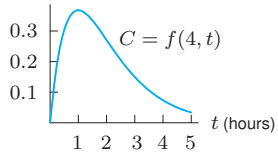
15 (a)



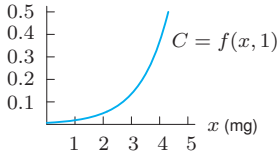
(b)



17 (a) C (mg per liter)

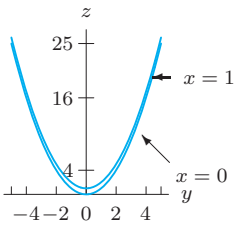


(b) C (mg per liter)

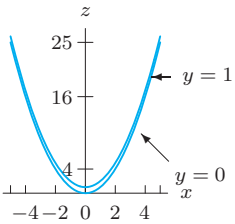


19 (a) is (IV)
(b) is (IX)
(c) is (VII)
(d) is (I)
(e) is (VIII)
(f) is (II)
(g) is (VI)
(h) is (III)
(i) is (V)

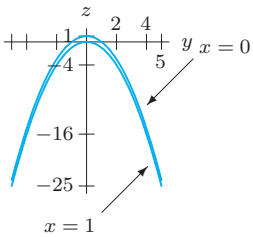
21 (a) (i)



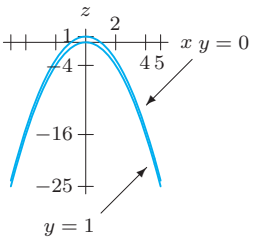
(ii)



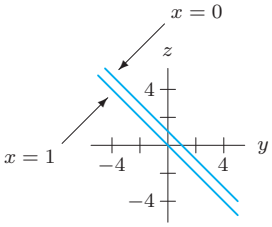
(b) (i)



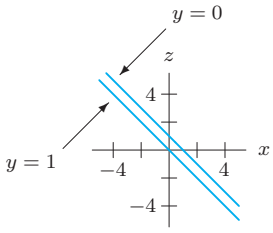
(ii)



(c) (i)

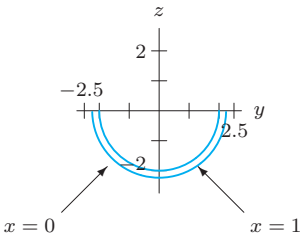


(ii)

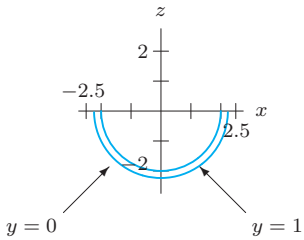


(d)

(i)

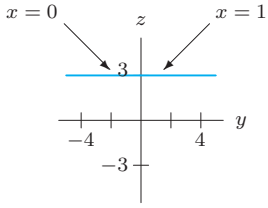


(ii)

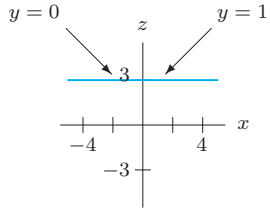


(e)

(i)



(ii)



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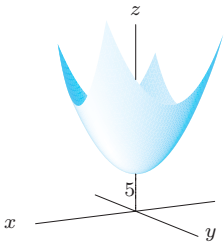

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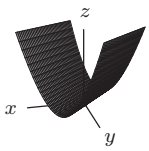

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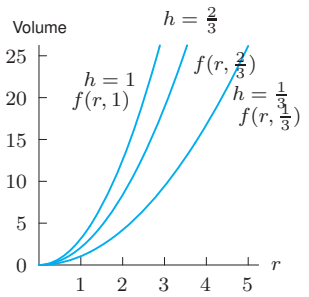
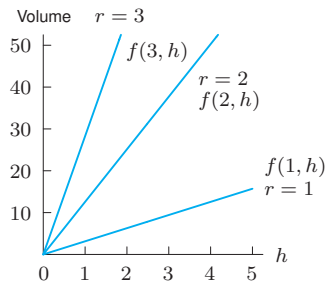
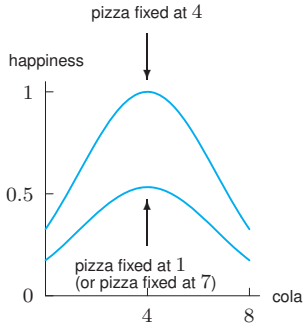


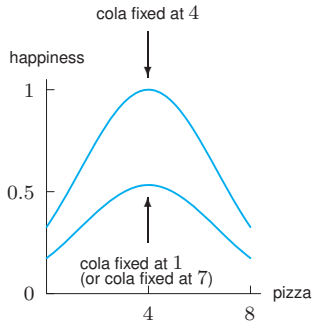


29 Cross-sections graph I:

(a)

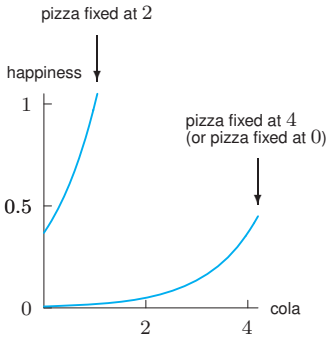


(b)

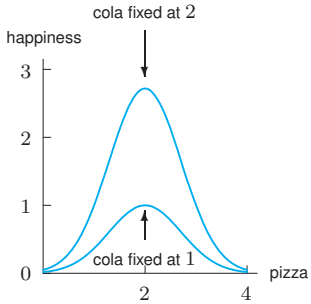


Cross-sections graph II:

(a)

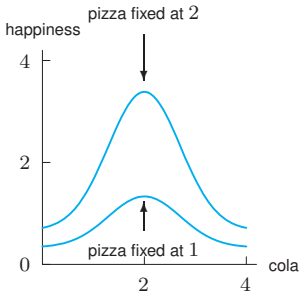


(b)

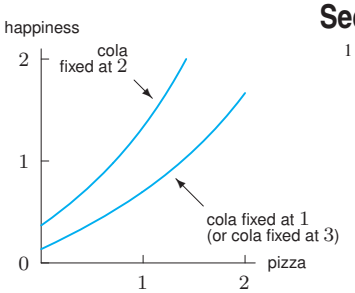


Cross-sections graph III:

(a)

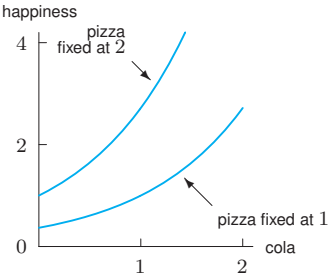


(b)

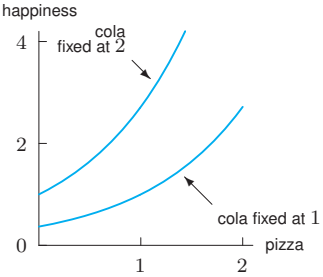


Cross-sections graph IV:

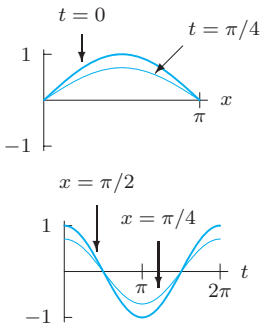
(a)



(b)



31 (a)



(b) $f = 0$; ends of string don't move

33 Cross-sections are lines parallel to y-axis

35 $f(x, y) = x^2 - 1$

37 False

39 True

41 True

43 True

45 True

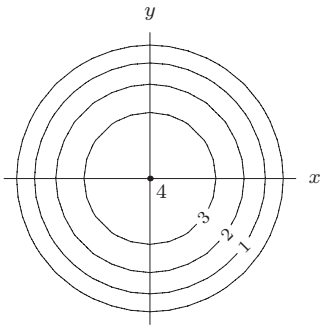
47 True

49 False

51 (c)

Section 12.3

1



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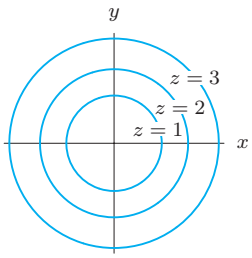
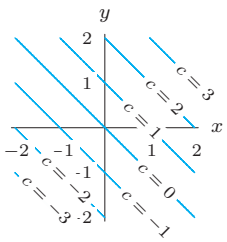
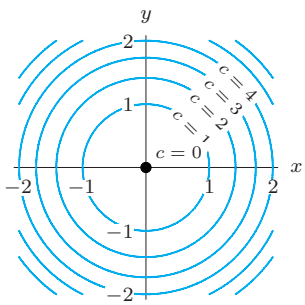


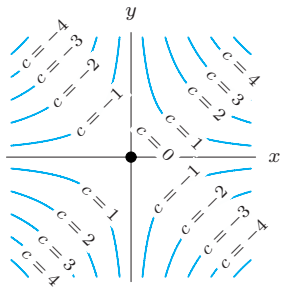
5 Contours evenly spaced



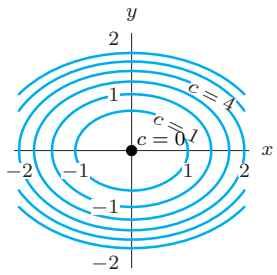
7



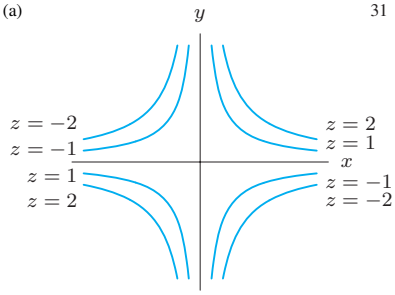
9



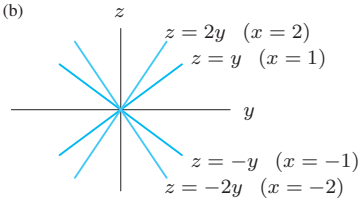
11



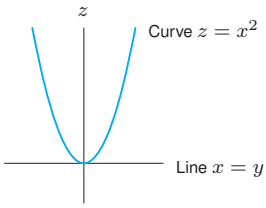
15 (a)



(b)



(c)

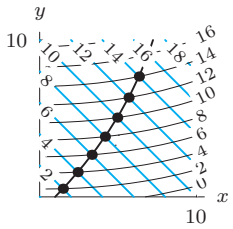


17 Table 12.5 matches (II)
Table 12.6 matches (III)
Table 12.7 matches (IV)
Table 12.8 matches (I)

19 Price: x; Advertising: y

21 (a) is III
(b) is VI
(c) is I
(d) is IV
(e) is II
(f) is V

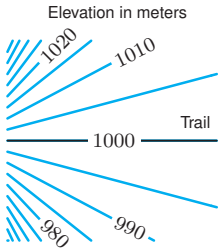
23



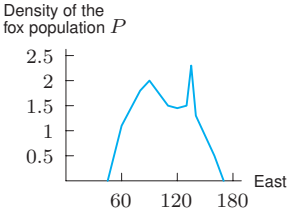
25 72°F; 76°F

27 (a) A
(b) B
(c) A

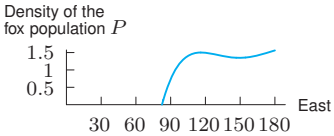
29



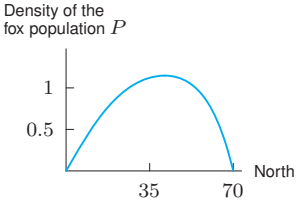
31 E-W, $N = 50$:



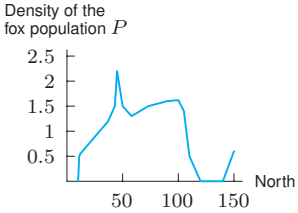
E-W, $N = 100$:



N-S, $E = 60$:



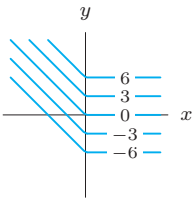
N-S, $E = 120$:



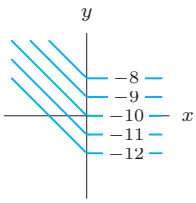
33 (a) (II) (E)
(b) (I) (D)
(c) (III) (G)

35 (a) (I) g
(II) f
(b) $0.2 < \alpha < 0.8$

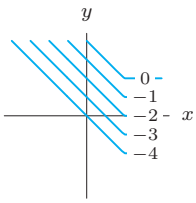
37 (a)



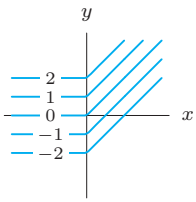
(b)



(c)



(d)



43 Contour diagram is curves in xy-plane

45 $f(x, y) = x^2$

47 Could not be true

49 Might be true

51 True

53 True

55 False

57 False

59 True

Section 12.4

1 -1.0

3 Not a linear function

5 Linear function

7 Linear

9 $z = \frac{4}{3}x - \frac{1}{2}y$

11 $z = -2y + 2$

13 $\Delta z = 0.4; \quad z = 2.4$

15 CDs cost \$8
DVDs cost \$12

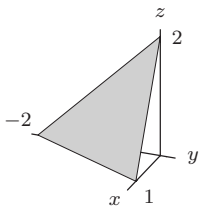
17 (a) Linear
(b) Linear
(c) Not linear

19 180 lb person at 8 mph
120 lb person at 10 mph

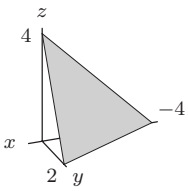
21 $g(x, y) = 3x + y$

23 $f(x, y) = 2x - 0.5y + 1$

25



27



31 (a) $7/\sqrt{29}$
(b) $-5/\sqrt{104}$

33 Contours of $f(x, y) = e^{x+y}$ parallel lines

35

$x \backslash y$	1	2	3
1	1	2	3
2	2	4	6
3	3	6	9

37 False

39 False

41 True

43 True

45 True

47 False

Section 12.5

1 (a) I
(b) II

3 $f(x, y) = \frac{1}{3}(5 - x - 2y)$

5 $f(x, y) = (1 - x^2 - y)^2$

7 Elliptical and hyperbolic paraboloid, plane

9 Hyperboloid of two sheets

11 Ellipsoid

13 Yes, $f(x, y) = (2x + 3y - 10)/5$

15 No

17 $f(x, y) = \sqrt{10 - x^2 - y^2}$
$g(x, y, z) = x^2 + y^2 + z^2 = 10$

19 (a) Parallel planes: $2x - 3y + z = c + 20$
(b) $f_z(0, 0, 0) = 1$
(c) $\vec{n} = 2\vec{i} - 3\vec{j} + \vec{k}$
(d) Yes; $-3°$F

21 Incr of A and r
Decr of t

23 $f(x, y) = 3\sqrt{1 - x^2 - y^2/4}$;
$g(x, y) = -3\sqrt{1 - x^2 - y^2/4}$

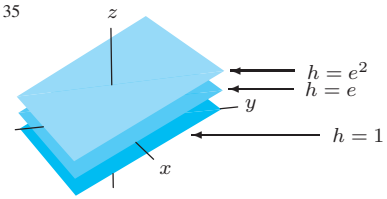
25 (a) Graph of f is graph of
$x^2 + y^2 + z^2 = 1, z \geq 0$
(b) $\sqrt{1 - x^2 - y^2} - z = 0$

27 $g(x, y, z) = 1 - \sqrt{x^2 + z^2} - y = 0$

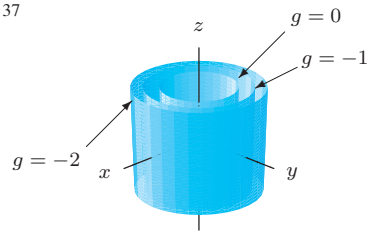
29 Ellipsoid

31 Parallel planes

35



37



39 Level surfaces hyperbolic cylinders

41 $f(x, y, z) = y + z$

43 $f(x, y, z) = (x + y + z)^2$

45 True

47 True

49 True

51 False

53 False

Section 12.6

1 Not continuous

3 Continuous

5 Not continuous

7 1

9 0

11 1

19 No

21 $c = 1$

23 (c) No

25 For quotient, need $g(a, b) \neq 0$

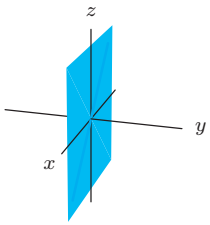
27 $f(x, y) = (x^2 + 2y^2)/(x^2 + y^2)$

29 $f(x, y) = 1/((x - 2)^2 + y^2)$

Chapter 12 Review

1 A, B, C

3

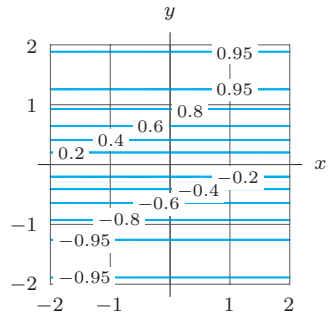


5 Not a function

7 IV

9 (a) is (I)
(b) is (IV)
(c) is (II)
(d) is (III)

11 $y = k$



13 $2x^2 + y^2 = k$

15 $(x-1)^2 + (y-2)^2 + (z-3)^2 = 25$

17 $(-2, 3, -6)$; 7

19 Linear

21 $f(x, y, z) = x^2 + y^2$

23 $z = f(x, y) = 4 - 2x - 4y/3$
$g(x, y, z) = (x/2) + (y/3) + (z/4) = 1$

25 $z = f(x, y) = -\sqrt{4 - (x-3)^2 - y^2}$
$g(x, y, z) = (x-3)^2 + y^2 + z^2 = 4$

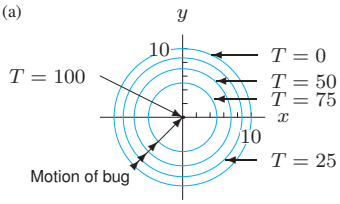
27 Cylinder

29 $g(x, y) = 200x + 100y$

31 Doubles production

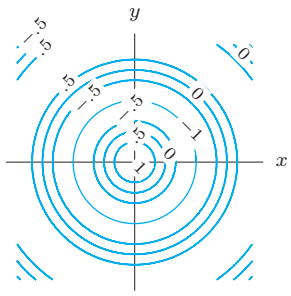
37 (a) No
(b) No
(c) No
(e) 0.5 of GPA

39 (a)

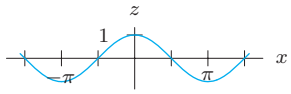


(b) Toward origin

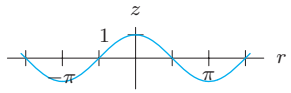
41 (a)



(b)



(c)



47 (a) $9 + 3x + 4y, 21 + 7x + 8y$
(b) $189 + 63x + 64y$

Section 13.1

1 $\vec{a} = \vec{i} + 3\vec{j}$
$\vec{b} = 3\vec{i} + 2\vec{j}$
$\vec{v} = -2\vec{i} - 2\vec{j}$
$\vec{w} = -\vec{i} + 2\vec{j}$

3 $-3\vec{i} - 4\vec{j}$

5 $\vec{a} = \vec{b} = \vec{c} = 3\vec{k}$
$\vec{d} = 2\vec{i} + 3\vec{k}$
$\vec{e} = \vec{j}$
$\vec{f} = -2\vec{i}$

7 $\vec{i} + 3\vec{j}$

9 $-4.5\vec{i} + 8\vec{j} + 0.5\vec{k}$

11 $-3\vec{i} - 12\vec{j} + 3\vec{k}$

13 $0.9\vec{i} + 0.2\vec{j} - 0.7\vec{k}$

15 $\sqrt{6}$

17 $\sqrt{11}$

19 5.6

21 $-6\vec{i} + 20\vec{j} + 13\vec{k}$

23 $21\vec{j}$

25 $2\sqrt{73}$

27 $0.6\vec{i} - 0.8\vec{k}$

29 $-\vec{i}/2 + \vec{j}/4 + \sqrt{11}\vec{k}/4$

31 $0, -10$

33 $-\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$.

35 (a) $\vec{u} = 8.2\vec{i} + 1.8\vec{j}$,
$\vec{v} = -8.2\vec{i} - 1.8\vec{j}$
(b) $\vec{v} = -\vec{u}$

37 (a) True
(b) Does not makes sense
(c) True
(d) False

39 (a) $t = 1$
(b) No t values
(c) Any t values

41 $-\vec{i}/2 + \sqrt{3}\vec{j}/2$
$-\vec{i}/2 - \sqrt{3}\vec{j}/2$

45 Opposite direction if $c < 0$, no direction if $c = 0$

47 $\vec{w} = \vec{0}$ so $\vec{v} + \vec{w} = \vec{v}$

49 $\vec{u} = \vec{i}, \vec{v} = (1/2)\vec{i} + (\sqrt{3}/2)\vec{j}$

51 False

53 True

55 False

57 False

59 False

Section 13.2

1 Scalar

3 Scalar

5 Vector

7 $-37.59\vec{i}, -13.68\vec{j}$

9 $21\vec{i} + 35\vec{j}$

11 (a) 8.64 km/hr
(b) 0.093 radian or about 5° off course

13 (a) $17.93\vec{i} - 7.07\vec{j}$
(b) 19.27 km/hr
(c) 21.52° south of east

15 48.3° east of north
744 km/hr

17 4.87° north of east
540.63 km/hr

19 $\vec{E} = 27.908\vec{i} + 1.823\vec{j}$
Speed 27.97 km/hr in direction 3.74° north of east

21 38.7° south of east

23 $-98.76\vec{i} + 18.94\vec{j} + 2998.31\vec{k}$
2998.31 newtons directly up

25 $0.4v\vec{i} + 0.7v\vec{j}$

27 $0.01875\vec{i} + .0125\vec{j} + .03125\vec{k}$

39 $\vec{i}$ or $\vec{j}$ components could be different

41 $\vec{F} = \vec{i} - 2\vec{j}$

43 Yes

45 Yes

47 Yes

Section 13.3

1 -14

3 14

5 -2

7 $28\vec{j} + 14\vec{k}$

9 185

11 $\vec{i} + 3\vec{j} + 2\vec{k}$ (multiples of)

13 $3\vec{i} + 4\vec{j} - \vec{k}$ (multiples of)

15 $3x - y + 4z = 6$

17 $x - y + z = 3$

19 $2x + 4y - 3z = 5$

21 $2x - 3y + 5z = -17$

23 $2\pi/3$ radians (120°)

25 $\pi/6$ radians (30°)

27 (a) $(2/\sqrt{13})\vec{i} + (3/\sqrt{13})\vec{j}$
(b) Multiples of $3\vec{i} - 2\vec{j}$

29 (a) $(21/5, 0, 0)$
(b) $(0, -21, 0)$; $(0, 0, 3)$ (for example)
(c) $\vec{n} = 5\vec{i} - \vec{j} + 7\vec{k}$ (for example)
(d) $21\vec{j} + 3\vec{k}$ (for example)

31 Possible answers:
(a) $2\vec{i} + 3\vec{j} - \vec{k}$
(b) $3\vec{i} - 2\vec{j}$

33 (a) $(\vec{a}, \vec{c})$ and $(\vec{b}, \vec{d})$
(b) None
(c) $(\vec{a}, \vec{d})$ and $(\vec{c}, \vec{d})$
(d) $(\vec{a}, \vec{b})$ and $(\vec{b}, \vec{c})$

35 Many answers possible
(a) $8\vec{i} + 6\vec{j}$
(b) $-3\vec{i} + 4\vec{j}$

37 (a) Increases
(b) Decreases

39 71.57°

41 (a) $\sqrt{2}$
(b) 0.32

43 (a) $\vec{F}_{\text{parallel}} = -0.168\vec{i} - 0.224\vec{j}$
(b) $\vec{F}_{\text{perp}} = 0.368\vec{i} - 0.276\vec{j}$
(c) $W = -1.4$

45 (a) $\vec{F}_{\text{parallel}} = \vec{0}$
(b) $\vec{F}_{\text{perp}} = \vec{F}$
(c) $W = 0$

47 (a) $\vec{F}_{\text{parallel}} = \vec{F}$

(b) $\vec{F}_{\text{perp}} = \vec{0}$
(c) $W = -50$

49 (a) $\vec{F}_{\text{parallel}} = 3.846\vec{i} - 0.769\vec{j}$
(b) $\vec{F}_{\text{perp}} = -3.846\vec{i} - 19.231\vec{j}$
(c) $W = 20$

51 (a) $\vec{F}_{\text{parallel}} = \vec{0}$
(b) $\vec{F}_{\text{perp}} = \vec{F}$
(c) $W = 0$

53 No

55 (a) 2.24 m/sec
(b) 1.89 m/sec

57 $10/\sqrt{30}$

59 Averages; weightings
85.65%; class average

71 Can't take dot product of a scalar and a vector

73 Normal vector is $2\vec{i} + 3\vec{j} - \vec{k}$

75 $f(x, y) = (-1/3)x + (-2/3)y$

77 True

79 False

81 True

83 False

85 True

Section 13.4

1 $-\vec{i}$

3 $-\vec{i} + \vec{j} + \vec{k}$

5 $\vec{i} + 3\vec{j} + 7\vec{k}$

7 $7\vec{i} + \vec{j} + 4\vec{k}$

9 $-2\vec{k}$

11 $\vec{i} - \vec{j}$

13 $\vec{v} \times \vec{w} = -6\vec{i} + 7\vec{j} + 8\vec{k}$
$\vec{w} \times \vec{v} = 6\vec{i} - 7\vec{j} - 8\vec{k}$
$\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$

15 $x - y - z = -3$

17 4

19 0

21 $4x + 26y + 14z = 0$

23 $3x - y - 2z = 0$

25 $4\vec{i} + 26\vec{j} + 14\vec{k}$

27 $4(x - 4) + 26(y - 5) + 14(z - 6) = 0$

29 (a) $\vec{u}$ and $-\vec{u}$ where
$\vec{u} = \frac{12}{13}\vec{i} - \frac{4}{13}\vec{j} - \frac{3}{13}\vec{k}$
(b) $\theta \approx 49.76°$
(c) $13/2$
(d) $13/\sqrt{29}$

31 Parallel to the z-axis

33 $\sqrt{38}/3$

35 (a) Increases force
(b) Ball moves down and to the left

37 (a) $\overrightarrow{P_1P_2} = \overrightarrow{P_3P_4} = 2\vec{i} + 4\vec{j} + 2\vec{k}$;
$\overrightarrow{P_1P_3} = \overrightarrow{P_2P_4} = 3\vec{i}$
(b) $\sqrt{180}$

51 $4\pi\vec{i}$

53 (a) $((u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2)^{1/2}$
(b) $|u_1v_2 - u_2v_1|$
(c) $m = (u_2v_3 - u_3v_2)/(u_2v_1 - u_1v_2)$,
$n = (u_3v_1 - u_1v_3)/(u_2v_1 - u_1v_2)$

55 Parallel, not perpendicular

57 $\vec{v} = (8\vec{i} - 6\vec{j})/5$

59 False

61 True

63 True

65 True

67 False

Chapter 13 Review

1 Scalar; -1

3 -1

5 $\vec{a} = -2\vec{j}$, $\vec{b} = 3\vec{i}$, $\vec{c} = \vec{i} + \vec{j}$,
$\vec{d} = 2\vec{j}$, $\vec{e} = \vec{i} - 2\vec{j}$, $\vec{f} = -3\vec{i} - \vec{j}$

7 $5\vec{i} + 30\vec{j}$

9 $3\sqrt{2}$

11 $3\vec{i} + 7\vec{j} - 4\vec{k}$

13 -3

15 $\vec{0}$

17 0

19 $\vec{0}$

21 $-5\vec{i} + 3\vec{j} + \vec{k}$ (multiples of)

23 (a) 4
(b) $-4\vec{i} - 11\vec{j} - 17\vec{k}$
(c) $3.64\vec{i} + 2.43\vec{j} - 2.43\vec{k}$
(d) $79.0°$
(e) 0.784.
(f) $2\vec{i} - 2\vec{j} + \vec{k}$ (many answers possible)
(g) $-4\vec{i} - 11\vec{j} - 17\vec{k}$.

25 $\pm(-\vec{i} + \vec{j} - 2\vec{k})/\sqrt{6}$

27 $\vec{n} = 4\vec{i} + 6\vec{k}$

29 $-3\vec{i} + 4\vec{j}$

31 $\vec{u}$ and $\vec{w}$; $\vec{v}$ and $\vec{q}$.

33 $\vec{F}_{\text{parallel}} = \vec{F}$
$\vec{F}_{\text{perp}} = \vec{0}$
$W = -10$

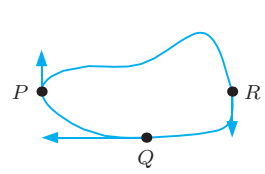
35 $\vec{F}_{\text{parallel}} = -(6/5)\vec{i} + (8/5)\vec{j}$
$\vec{F}_{\text{perp}} = (16/5)\vec{i} + (12/5)\vec{j}$
$W = -10$

37 $\vec{F}_{\text{parallel}} = 2\vec{j}$
$\vec{F}_{\text{perp}} = 5\vec{i}$
$W = 6$

39 (a) True
(b) False
(c) False
(d) True
(e) True
(f) False

41 548.6 km/hr

43



45 Parallel:
$3\vec{i} + \sqrt{3}\vec{j}$ and $\sqrt{3}\vec{i} + \vec{j}$
Perpendicular:
$\sqrt{3}\vec{i} + \vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$
$3\vec{i} + \sqrt{3}\vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$

47 2, 8

49 $2x - 3y + 7z = 19$

51 $\sqrt{6}/2$

53 (a) 1.5
(b) $y = 1$

55 $9x - 16y + 12z = 5$
0.23

57 $7.0710\vec{i} + 2.5882\vec{j} + 6.580\vec{k}$

59 228.43 newtons
85.5° south of east

61 $|ywr - vzr + zus - xws + xvt - yut|$

63 (b) $(1, 1/\sqrt{3}, \sqrt{6}/6)$
(c) $109.471°$

Section 14.1

1 $f_x(3, 2) \approx -2/5$; $f_y(3, 2) \approx 3/5$

3 $-0.0493, -0.3660$
$-0.0501, -0.3629$

5 $\partial P/\partial t$:
dollars/month
Rate of change in payments with time
negative
$\partial P/\partial r$:
dollars/percentage point
Rate of change in payments with interest rate
positive

7 (a) Payment \$376.59/mo at 1% for 24 mos
(b) 4.7¢ extra/mo for \$1 increase
(c) Approx \$44.83 increase for 1% interest increase

9 (a) Negative
(b) Positive

11 $f_x > 0$, $f_y < 0$

13 $f_x < 0$, $f_y > 0$

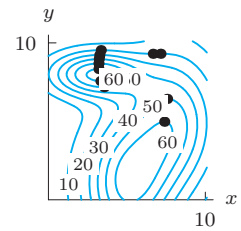
15 $f_T(5, 20) \approx 1.2°\text{F}/°\text{F}$

17 Positive, Negative, 10, 2, -4

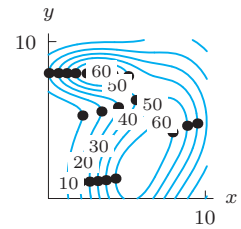
19 $\partial Q/\partial b < 0$
$\partial Q/\partial c > 0$

21 -1.5 and -1.22

23 (a)



(b)



25 (a) Negative
(b) Positive

27 (a) 2.5, 0.02
(b) 3.33, 0.02
(c) 3.33, 0.02

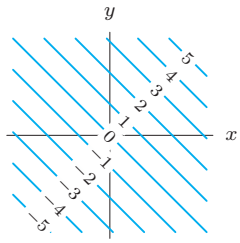
29 (i)(c); (ii)(a)

31 -2.5

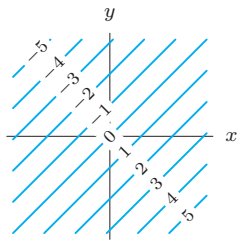
33

T (°C) \ w (gm/m^3)	0.1	0.2	0.3
10	1300	900	1200
20	800	800	900
30	800	700	800

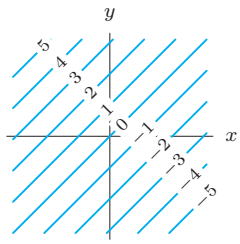
35 (a)



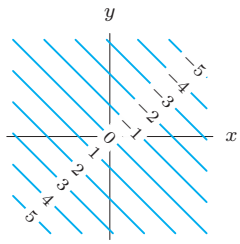
(b)



(c)



(d)



37 $f_y = 0$

39 $f(x, y) = 4x - y$

41 False

43 False

45 True

47 False

49 False

Section 14.2

1 (a) 7.01
(b) 7

3 $f_x(1, 2) = 15$,
$f_y(1, 2) = -5$

5 $\partial z/\partial x = \frac{14x+7}{(x^2+x-y)^{-6}}$
$\partial z/\partial y = -7(x^2 + x - y)^6$

7 $f_x = 0.6/x$
$f_y = 0.4/y$

9 $2xy + 10x^4y$

11 $-2\pi r/T^2$

13 $e^{\sqrt{xy}}(1 + \sqrt{xy}/2)$

15 g

17 $(a + b)/2$

19 $2B/u_0$

21 $2mv/r$

23 Gm_1/r^2

25 $-e^{-x^2/a^2}(a^2 - 2x^2)/a^4$

27 $v_0 + at$

29 $-\pi r^{3/2}/(M\sqrt{GM})$

31 $(15x^2y - 3y^2)\cos(5x^3y - 3xy^2)$

33 $\partial F/\partial L = \frac{3}{2}\sqrt{K/L}$

35 $\epsilon_0 E$

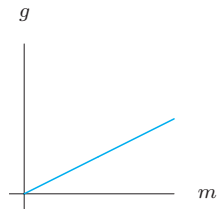
37 $c\gamma a_1 b_1 K^{b_1-1}(a_1K^{b_1} + a_2L^{b_2})^{\gamma-1}$

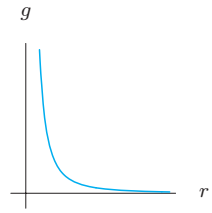
39 13.6

41 (a) 3.3, 2.5
(b) 4.1, 2.1
(c) 4, 2

43 (a) $a < 3$
(b) Up

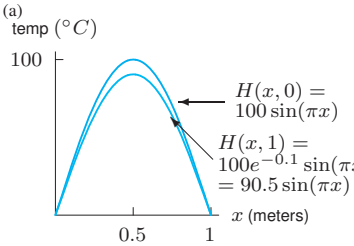
45 (a) $\partial g/\partial m = G/r^2$
$\partial g/\partial r = -2Gm/r^3$
(b)





47 (a) $c^2((1 - v^2/c^2)^{-1/2} - 1)$; positive
(b) $mv(1 - v^2/c^2)^{-3/2}$; positive

49 (a)



(b) $254.2e^{-0.1t}$ °C/m
(c) $-254.2e^{-0.1t}$ °C/m
(d) $-10e^{-0.1t}\sin(\pi x)$ °C/sec

51 $f(x, y) = x^4y^2 - 3xy^4 + C$

53 Derivative is limit, not a difference quotient
Counterexample: $f(x, y) = x^2 + y^2$

55 $f(x, y) = xy, g(x, y) = xy + y^2$

57 True

59 True

61 False

63 False

65 (d)

Section 14.3

1 $z = ex$

3 $z = 6y - 9$

5 $z = -4 + 2x + 4y$

7 $z = -36x - 24y + 148$

9 $df = y\cos(xy)\,dx + x\cos(xy)\,dy$

11 $dz = -e^{-x}\cos(y)dx - e^{-x}\sin(y)dy$

13 $dg = 4\,dx$

15 $dP \approx 2.395\,dK + 0.008\,dL$

17 (a) Dollars/Square foot
(b) Larger plots at same distance \$3/ft^2 more
(c) Dollars/Foot
(d) Farther from beach but same area \$2/ft less
(e) 998 ft^2

19 (b) $f(x, y) \approx$
$0.3345 - 0.33(x - 1) - 0.15(y - 2)$
(c) $f(x, y) \approx 0.3345 -$
$0.3345(x - 1) - 0.1531(y - 2)$

21 376

23 $df = \frac{1}{3}dx + 2dy$
$f(1.04, 1.98) \approx 2.973$

25 136.09°C

27 $P(r, L) \approx$
$80 + 2.5(r - 8) + 0.02(L - 4000)$
$P(r, L) \approx$
$120 + 3.33(r - 8) + 0.02(L - 6000)$
$P(r, L) \approx$
$160 + 3.33(r - 13) + 0.02(L - 7000)$

31 (a) $nRT/(V - nb) - n^2a/V^2$
(b) $\Delta P \approx (nR/(V_0 - nb))\Delta T +$
$(2n^2a/V_0^3 - nRT_0/((V_0 - nb)^2))\Delta V$

33 (a) $d\rho = -\beta\rho\,dT$
(b) $0.00015, \beta \approx 0.0005$

35 $-43200\Delta t$
Slow if $\Delta t > 0$; fast if $\Delta t < 0$

37 $z = f(3, 4) + f_x(3, 4)(x - 3) +$
$f_y(3, 4)(y - 4)$

39 Equation not linear

41 sphere of radius 3 centered at the origin

43 True

45 False

47 True

49 False

Section 14.4

1 $(\frac{15}{2}x^4)\vec{i} - (\frac{24}{7}y^5)\vec{j}$

3 $2m\vec{i} + 2n\vec{j}$

5 $\left(\frac{5\alpha}{\sqrt{5\alpha^2+\beta}}\right)\vec{i} + \left(\frac{1}{2\sqrt{5\alpha^2+\beta}}\right)\vec{j}$

7 $\nabla z = e^y\vec{i} + e^y(1 + x + y)\vec{j}$

9 $\sin\theta\vec{i} + r\cos\theta\vec{j}$

11 $\nabla z = \frac{1}{y}\cos\left(\frac{x}{y}\right)\vec{i} - \frac{x}{y^2}\cos\left(\frac{x}{y}\right)\vec{j}$

13 $\left(\frac{-12\beta}{(2\alpha-3\beta)^2}\right)\vec{i} + \left(\frac{12\alpha}{(2\alpha-3\beta)^2}\right)\vec{j}$

15 $60\vec{i} + 85\vec{j}$

17 $10\pi\vec{i} + 4\pi\vec{j}$

19 $(\pi/2)^{1/2}\,\vec{i}$

21 $\frac{1}{100}(2\vec{i} - 6\vec{j})$

23 $-46/5$

25 $22/5$

27 $84/5$

29 $(2x + 3e^y)dx + 3xe^y dy$

31 $(x+1)ye^x\vec{i} + xe^x\vec{j}$

33 Negative

35 Positive

37 Positive

39 $-\vec{j}$

41 $\vec{j}$

43 $-\vec{i} - \vec{j}$

45 $\vec{i} - \vec{j}$

47 (a) $(3.96838, 5.09487)$
(b) 0.1052
(c) $1/(3\sqrt{10})$

49 4.4

51 (a) $-16\vec{i} + 12\vec{j}$
(b) $16\vec{i} - 12\vec{j}$
(c) $12\vec{i} + 16\vec{j}$; answers may vary

53 (a) -3.268
(b) -4.919

55 (a) Should be number
(b) $11/5$

57 -0.5; better estimate is -1.35

59 -1.1; better estimate is -1.8

61 $4\vec{i} + 6\vec{j}$; $4(x-2) + 6(y-3) = 0$

63 $-4\vec{i} + \vec{j}$; $-4(x-2) + (y-3) = 0$

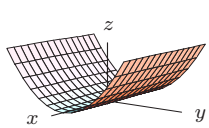
65 (a) Negative
(b) Negative

67 (a) Negative
(b) Positive
(c) Positive
(d) Negative

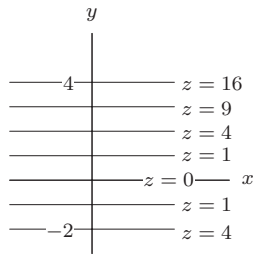
69 Yes

71 Yes

73 (a)



(b)



(c) $\vec{j}$

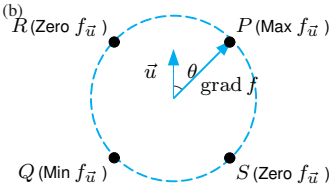
75 (a) Circles centered at P
(b) away from P
(c) 1

77 $(3\sqrt{5} - 2\sqrt{2})\vec{i} + (4\sqrt{2} - 3\sqrt{5})\vec{j}$

79 $4\sqrt{2}$,
$6\vec{i} + 2\vec{j}$

81 $5/\sqrt{2}$

83 (a) P, Q
(b)

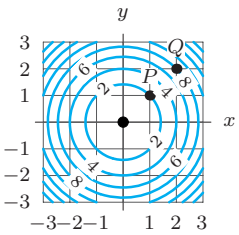


(c) $\|\text{grad } f\|$
$f_{\vec{u}} = \|\text{grad } f\|\cos\theta$

89 $f_{\vec{u}}\ (0,0)$ is scalar, not vector

91 Closer contours give longer gradients

93



95 False

97 False

99 True

101 False

103 True

105 True

107 True

109 True

111 False

Section 14.5

1 $2x\vec{i}$

3 $e^x e^y e^z(\vec{i} + \vec{j} + \vec{k})$

5 $\frac{-2xyz^2}{(1+x^2)^2}\vec{i} + \frac{z^2}{1+x^2}\vec{j} + \frac{2yz}{1+x^2}\vec{k}$

7 $(x\vec{i} + y\vec{j} + z\vec{k})/\sqrt{x^2+y^2+z^2}$

9 $y\vec{i} + x\vec{j} + e^z\cos(e^z)\,\vec{k}$

11 $e^p\vec{i} + (1/q)\vec{j} + 2re^{r^2}\vec{k}$

13 $\vec{0}$

15 $6\vec{i} + 4\vec{j} - 4\vec{k}$

17 $-\pi\vec{i} - \pi\vec{k}$

19 $9/\sqrt{3}$

21 $-1/\sqrt{2}$

23 $-\sqrt{77/2}$

25 $-2\vec{i} - 2\vec{j} + 4\vec{k}$;
$-2(x+1) - 2(y-1) + 4(z-2) = 0$

27 $2\vec{j} - 4\vec{k}$; $2(y-1) - 4(z-2) = 0$

29 $-2\vec{i} + \vec{k}$; $-2(x+1) + (z-2) = 0$

31 (a) 0
(b) $24/\sqrt{19}$

33 (a) $(2x - yz)\vec{i} + (2y - xz)\vec{j} - xy\vec{k}$
(b) $x + 4y + 6z = 20$

35 $z = 12 - 4x - 2y$

37 $6(x-1) + 3(y-2) + 2(z-1) = 0$

39 $2x + 3y + 2z = 17$

41 $z = 2x + y + 3$

43 $x + 4y + 10z = 18$

45 (a) is (III); (b) is (I); (c) is (IV); (d) is (II)

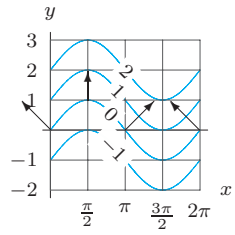
47 $22/\sqrt{14}$

49 (a) Spheres centered at the origin
(b) $2x\sin(x^2+y^2+z^2)\vec{i} + 2y\sin(x^2+y^2+z^2)\vec{j} + 2z\sin(x^2+y^2+z^2)\vec{k}$
(c) $0, 180°$

51 (a) $-\vec{j}$,
$\frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} + \frac{2}{3}\vec{k}$
(b) $y = 0$; $2x - y + 2z = 3$
(c) $x = y = 0$ and $z \neq 0$

53 (a) $1.2x + 1.6y + 6z = 8$
(b) Yes; $\pm(0.8, 0.6, 1)$

55 (a)



(b) Valley

57 (a) $-4\vec{i} + 4\vec{j} + 2\vec{k}$
(b) 24 mg/cm^3/sec

59 $f_{\vec{u}\,3} < f_{\vec{u}\,2} < 0 < f_{\vec{u}\,4} < f_{\vec{u}\,1}$

61 (a) Spheres centered at origin
(b) $(-2x\vec{i} - 2y\vec{j} - 2z\vec{k})e^{-(x^2+y^2+z^2)}$
(c) $-3\sqrt{2}e^{-1}$ degrees/sec

63 (a) is (V); (b) is (IV); (c) is (V)

71 $f_x(0,0,0)x + f_y(0,0,0)y + f_z(0,0,0)z = 0$

73 $f(x,y,z) = 2x + 3y + 4z + 100$

75 False

77 False

79 (a) °C per meter
(b) °C per second
(c) °C per second

Section 14.6

1 $\dfrac{dz}{dt} = e^{-t}\sin(t)(2\cos t - \sin t)$

3 $2\cos\left(\frac{2t}{1-t^2}\right)\frac{1+t^2}{(1-t^2)^2}$

5 $2e^{1-t^2}(1-2t^2)$

7 $\dfrac{\partial z}{\partial u} = \dfrac{1}{vu}\cos\left(\dfrac{\ln u}{v}\right)$
$\dfrac{\partial z}{\partial v} = -\dfrac{\ln u}{v^2}\cos\left(\dfrac{\ln u}{v}\right)$

9 $\dfrac{\partial z}{\partial u} = \dfrac{e^v}{u}$
$\dfrac{\partial z}{\partial v} = e^v \ln u$

11 $\dfrac{\partial z}{\partial u} = 2ue^{(u^2-v^2)}(1+u^2+v^2)$
$\dfrac{\partial z}{\partial v} = 2ve^{(u^2-v^2)}(1-u^2-v^2)$

13 $\dfrac{\partial z}{\partial u} =$
$(e^{-v\cos u} - v(\cos u)e^{-u\sin v})\sin v$
$-(-u(\sin v)e^{-v\cos u} + e^{-u\sin v})v\sin u$
$\dfrac{\partial z}{\partial v} =$
$(e^{-v\cos u} - v(\cos u)e^{-u\sin v})u\cos v$
$+(-u(\sin v)e^{-v\cos u} + e^{-u\sin v})\cos u$

15 $\dfrac{\partial z}{\partial u} = \dfrac{-2uv^2}{u^4+v^4}$
$\dfrac{\partial z}{\partial v} = \dfrac{2vu^2}{u^4+v^4}$

17 $-2\rho\cos 2\phi$, 0

19 401.1

21 -5 pascal/hour

23 -0.6

25 Three

27 $\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$

29 $b\cdot e + d\cdot p$

35 $b\cdot e + d\cdot p$

37 (a) $\dfrac{\partial z}{\partial r} = \cos\theta\dfrac{\partial z}{\partial x} + \sin\theta\dfrac{\partial z}{\partial y}$
$\dfrac{\partial z}{\partial \theta} = r(\cos\theta\dfrac{\partial z}{\partial y} - \sin\theta\dfrac{\partial z}{\partial x})$
(b) $\dfrac{\partial z}{\partial y} = \sin\theta\dfrac{\partial z}{\partial r} + \dfrac{\cos\theta}{r}\dfrac{\partial z}{\partial \theta}$
$\dfrac{\partial z}{\partial x} = \cos\theta\dfrac{\partial z}{\partial r} - \dfrac{\sin\theta}{r}\dfrac{\partial z}{\partial \theta}$

39 $(\frac{\partial U_3}{\partial P})_V$

41 $(\frac{\partial U}{\partial T})_V = 7/2$
$(\frac{\partial U}{\partial V})_T = 11/4$

45 $\int_0^b F_u(x,y)\,dy$

49 C_R and C_T must be evaluated at $(5,1)$

51 $g(t) = 3+t$, $h(t) = t$

53 $z = x+y$, $x = e^t$ and $y = t^2$, many other answers are possible

55 $z = x+y$, $x = t$ and $y = t$

Section 14.7

1 $f_{xx} = 2$
$f_{yy} = 2$
$f_{yx} = 2$
$f_{xy} = 2$

3 $f_{xx} = 6y$
$f_{xy} = 6x + 15y^2$
$f_{yx} = 6x + 15y^2$
$f_{yy} = 30xy$

5 $f_{xx} = 0$
$f_{yx} = e^y = f_{xy}$
$f_{yy} = e^y(x+2+y)$

7 $f_{xx} = -\left(\sin\left(\frac{x}{y}\right)\right)\left(\frac{1}{y^2}\right)$
$f_{xy} = -\left(\sin\left(\frac{x}{y}\right)\right)\left(\frac{-x}{y^2}\right)\left(\frac{1}{y}\right)$
$+\left(\cos\left(\frac{x}{y}\right)\right)\left(\frac{-1}{y^2}\right) = f_{yx}$
$f_{yy} = -\left(\sin\left(\frac{x}{y}\right)\right)\left(\frac{-x}{y^2}\right)^2$
$+\left(\cos\left(\frac{x}{y}\right)\right)\left(\frac{2x}{y^3}\right)$

9 $f_{xx} = 30xy^2 + 18$
$f_{xy} = 30x^2y - 21y^2$
$f_{yx} = 30x^2y - 21y^2$
$f_{yy} = 10x^3 - 42xy$

11 $f_{xx} = -12\sin 2x\cos 5y$
$f_{xy} = -30\cos 2x\sin 5y$
$f_{yx} = -30\cos 2x\sin 5y$
$f_{yy} = -75\sin 2x\cos 5y$

13 $Q(x,y) = 1 + 2x - 2y + x^2 - 2xy + y^2$

15 $Q(x,y) = 1 + x + x^2/2 - y^2/2$

17 $Q(x,y) = 1 - x^2/2 - 3xy - (9/2)y^2$

19 $Q(x,y) = -y + x^2 - y^2/2$

21 $1 + x - y/2 - x^2/2 + xy/2 - y^2/8$

23 (a) Negative
(b) Zero
(c) Negative
(d) Zero
(e) Zero

25 (a) Positive
(b) Zero
(c) Positive
(d) Zero
(e) Zero

27 (a) Zero
(b) Negative
(c) Zero
(d) Negative
(e) Zero

29 (a) Positive
(b) Positive
(c) Zero
(d) Zero
(e) Zero

31 (a) Positive
(b) Negative
(c) Negative
(d) Negative
(e) Positive

33 $L(x,y) = y$
$Q(x,y) = y + 2(x-1)y$
$L(0.9, 0.2) = 0.2$
$Q(0.9, 0.2) = 0.16$
$f(0.9, 0.2) = 0.162$

39 $a = -b^2$

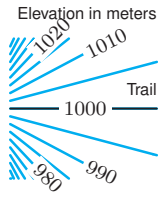
41 Positive, negative

43 (a) $z_{yx} = 4y$
(b) $z_{xyx} = 0$
(c) $z_{xyy} = 4$

45 $d = e = f = 0$

47 $d = 0$, $e > 0$, $f < 0$

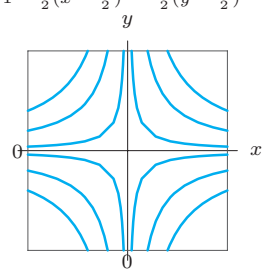
49 (a)

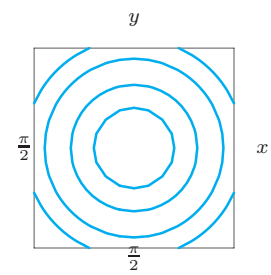


(b) $\partial h/\partial x = 0$, $\partial h/\partial y > 0$, $(\partial^2 h)/(\partial x\partial y) < 0$
(c) $(\partial^2 h)/(\partial x\partial y)$

51 (a) A
(b) B

53 (a) xy
$1 - \frac{1}{2}(x - \frac{\pi}{2})^2 - \frac{1}{2}(y - \frac{\pi}{2})^2$
(b)





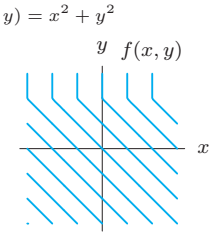
55 (a) (i) Dollars/Year, negative, (ii) Dollars/Dollar, positive
(b) $\partial^2 P/(\partial A\partial C) < 0$
(c) eCA

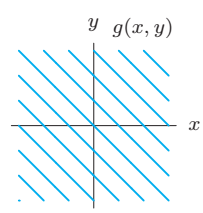
57 (a) $1 + x/2 + y$
(b) $1 + x/2 + y - x^2/8 - xy/2 - y^2/2$

61 Counterexample: $f(x,y) = x^3 + y^4$

63 $f(x,y) = x^2 + y^2$

65





Section 14.8

1 $(0,0)$

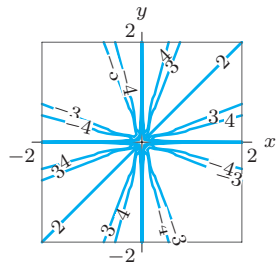
3 x-axis and y-axis

5 None

7 None

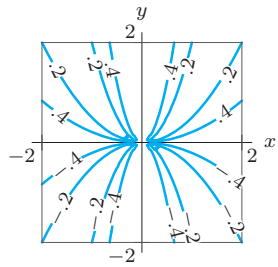
9 $(1, 2)$

11 (a)



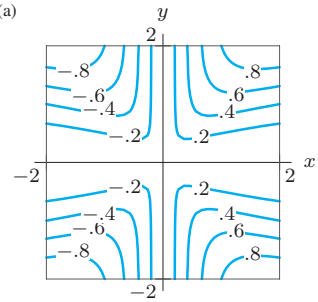
(b) No
(c) No
(d) No
(e) Exist, not continuous

13 (a)



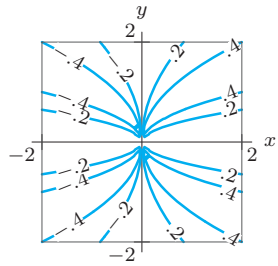
(b) Yes
(c) Yes
(d) No
(e) Exist, not continuous

15 (a)



(b) Yes
(d) No
(f) No

17 (a)



(c) No, no

19 (a) $f_x(x, y) = (x^4y + 4x^2y^3 - y^5)/(x^2 + y^2)^2$
$f_y(x, y) = (x^5 - 4x^3y^2 - xy^4)/(x^2 + y^2)^2$
(c) Yes
(d) Yes

21 Counterexample: $\sqrt{x^2 + y^2}$

23 $f(x, y) = \sqrt{x^2 + y^2}$

25 (a) Differentiable
(b) Not differentiable
(c) Not differentiable
(d) Differentiable

Chapter 14 Review

1 Vector; $3e^{-1}\vec{i} - \frac{1}{2}e^{-1}\vec{j}$

3 Vector;$-(\sin x)e^y\vec{i} + (\cos x)e^y\vec{j} + \vec{k}$

5 $f_x = 2xy + 3x^2 - 7y^6$
$f_y = x^2 - 42xy^5$

7 $\pi/\sqrt{lg}$

9 $f_x = \frac{2xy^3}{(x^2+y^2)^2}$, $f_y = \frac{x^4-x^2y^2}{(x^2+y^2)^2}$

11 $\partial f/\partial p = (1/q)e^{p/q}$
$\partial f/\partial q = -(p/q^2)e^{p/q}$

13 $f_N = c\alpha N^{\alpha-1}V^{\beta}$

15 $x/\left(2\sqrt{\omega x}\cos^2\left(\sqrt{\omega x}\right)\right)$

17 $\dfrac{270x^3y^7 - 168x^2y^6 - 15xy^2 + 16y}{(15xy - 8)^2}$

19 $\pi xy/\sqrt{2\pi xyw - 13x^7y^3v}$

21 $\frac{7}{2}\left(\frac{w-1}{x^2yw - xy^3w^7}\right)^{-9/2}$
$\left(\frac{x^2y + 6xy^3w^7 - 7xy^3w^6}{(w-1)^2}\right)$

23 $-1/(4\pi L\sqrt{LC})$

25 $u_{xx} = e^x \sin y, \quad u_{yy} = -e^x \sin y$

27 $f_{xxy} = f_{yxx} = 2\cos(x - 2y)$

29 $2x\vec{i} + (2y + 3y^2)\vec{j}$

31 $-((1/x)\vec{i} + (1/y)\vec{j} + (1/z)\vec{k})/(xyz)$

33 $\nabla z = 2x\cos(x^2 + y^2)\vec{i}$
$+ 2y\cos(x^2 + y^2)\vec{j}$

35 $\cos(x^2 + y^2 + z^2)\left(2x\vec{i} + 2y\vec{j} + 2z\vec{k}\right)$

37 $-\frac{(t^2-2t+4)}{(2s\sqrt{s})}\vec{i} + \frac{(2t-2)}{\sqrt{s}}\vec{j}$

39 $y[\cos(xy) - \sin(xy)]\vec{i}$
$+ x[\cos(xy) - \sin(xy)]\vec{j}$

41 $2\vec{i} + \vec{k}$

43 -1

45 0

47 $2/\sqrt{3}$

49 $5\vec{i} + 4\vec{j} + 3\vec{k}$

51 $-4x - 3y + 4z = 9$

53 $x + y + z = 3$

55 $\cos t \sin(\cos t) - \sin^2 t \cos(\cos t)$

57 $100t^3$

59 $3/t + 2t/(t^2 + 1)$

61 $Q(x, y) = 2 + 6x + y + 6x^2 + 3xy$

63 $Q(x, y) = 1 + (x - 3) - \frac{1}{2}(y - 5)$
$-\frac{1}{2}(x-3)^2 + \frac{1}{2}(x-3)(y-5) - \frac{1}{8}(y-5)^2$

65 (a) $2x - 4y + az = a - 2$
(b) $a = 2$

67 (a) Q, R
(b) Q, P
(c) P, Q, R, S
(d) None

71 $F = 684$ newtons,
$\partial F/\partial m = 9.77$ newtons/kg,
$\partial F/\partial r = -0.000214$ newtons/meter

73 (a) 20 hours per day
(b) 18.615 hours per day

75 (a) P, S
(b) R, S
(c) P, Q, R, S
(d) None

77 0.3

79 0.8

81 0

83 (a) $-5\sqrt{2}/2$
(b) $4\vec{i} + \vec{j}$

85 (a) 98.387 ft/mile
(b) 295.161 ft/hour

87 (a) $-4e^{-81}$ °C/meter
(b) $-40e^{-81}$ °C/sec
(c) $\sqrt{932}e^{-81}$ °C/meter

89 $-2x\vec{i} - 2y\vec{j}$

91 Yes

95 (a) $F_u(x, y, 3)$
(b) $F_w(3, y, x)$
(c) $F_u(x, y, x) + F_w(x, y, x)$
(d) $F_u(x, y, xy) + yF_w(x, y, xy)$

97 $dP \approx 47.6\,dL + 17.8\,dK$

101 (a) $-3\vec{i} + 4\vec{j} - \vec{k}$
(b) $-3\vec{i} + 4\vec{j}$

105 15°C/minute

107 Approx 7.5 at $(1.94, 1.08)$

109 $x - y$

111 (a) Negative, positive,
Up if positive, down if negative
(b) $\pi < t < 2\pi$
(c) $0 < x < 3\pi/2$ and
$0 < t < \pi/2$ or $3\pi/2 < t < 5\pi/2$

113 (a) $A_0 + A_1 + 2A_2 + A_3 + 2A_4 + 4A_5 + (A_1 + 2A_3 + 2A_4)(x - 1) + (A_2 + A_4 + 4A_5)(y - 2), 1 + B_1t, 2 + C_1t$
(b) $A_1B_1 + 2A_3B_1 + 2A_4B_1 + A_2C_1 + A_4C_1 + 4A_5C_1$

115 $\frac{\partial w}{\partial x}\frac{\partial x}{\partial u}\frac{du}{dt} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u}\frac{du}{dt} + \frac{\partial w}{\partial x}\frac{\partial x}{\partial v}\frac{dv}{dt} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v}\frac{dv}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$

Section 15.1

1 (I) and (V) Local maximum, (II) and (VI) Local minimum, (III) and (IV) Saddle point

3 (a) None
(b) E, G
(c) D, F

5 Local minimum

7 Local maximum

9 Local max: $(4, 2)$

11 Local max: $(1, 5)$

13 Saddle point: $(0, 0)$
Saddle point: $(2, 0)$
Local min: $(1, 0.25)$

15 Saddle pts: $(1, -1), (-1, 1)$
Local max: $(-1, -1)$
Local min: $(1, 1)$

17 Local max: $(-1, 0)$

Saddle pts: $(1, 0), (-1, 4)$
Local min: $(1, 4)$

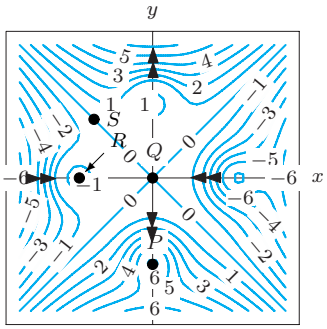
19 Saddle point: $(0, 0)$
Local max: $(1, 1), (-1, -1)$

21 $A = -2, B = 21$

23 (a) (a, b)
(b) $a = -1, b = 5$
(c) Local maximum

25 (a) Local maximum
(b) Saddle point
(c) Local minimum
(d) None of these

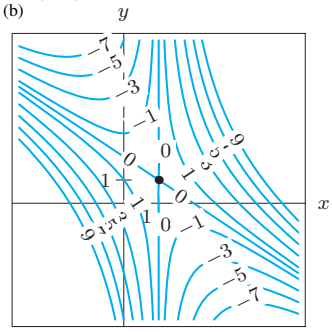
27



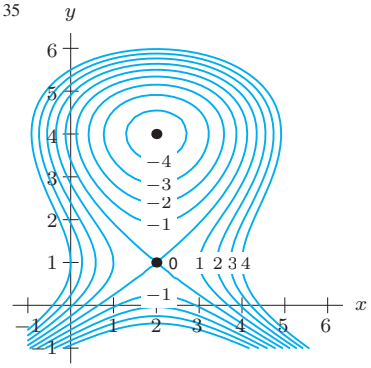
29 $(k\pi, l\pi)$,
for $k = 0, \pm 1, \pm 2, \cdots$,
$l = 0, \pm 1, \pm 2, \cdots$
$(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$,
for $k = 0, \pm 1, \pm 2, \cdots$,
$l = 0, \pm 1, \pm 2, \cdots$
$(k\pi, l\pi), k = 0, \pm 1, \pm 2, \cdots$,
$l = 0, \pm 1, \pm 2, \cdots$ are saddle points
If k and l are both even or k and l are both odd, then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local max
If k is even but l is odd or k is odd but l is even, then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local min

31 $y = 0, \pm 2\pi, \pm 4\pi, \ldots$ Local minima

33 (a) (a, b) is a saddle point.
(b)

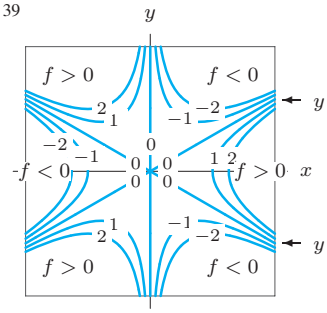


35



37 (a) $(0, 0)$
(b) $D = -24x^2$
(c) Saddle point

39



41 $(1, 3)$ could be saddle point

43 Can be saddle if f_{xy} large

45 $f(x, y) = 4 - (x - 2)^2 - (y + 3)^2$

47 False

49 True

51 False

53 True

55 False

Section 15.2

1 Mississippi:
$87 - 88$ (max), $83 - 87$ (min)
Alabama:
$88 - 89$ (max), $83 - 87$ (min)
Pennsylvania:
$89 - 90$ (max), 80 (min)
New York:
$81 - 84$ (max), $74 - 76$ (min)
California:
$100 - 101$ (max), $65 - 68$ (min)
Arizona:
$102 - 107$ (max), $85 - 87$ (min)
Massachusetts:
$81 - 84$ (max), 70 (min)

3 Max: 30.5 at $(0, 0)$
Min: 20.5 at $(2.5, 5)$

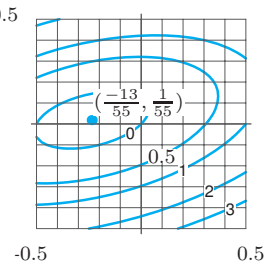
5 Min $= 0$ at $(0, 0)$
(not on boundary)
Max $= 2$ at $(1, 1), (1, -1)$, $(-1, -1)$ and $(-1, 1)$
(on boundary)

7 max$= 1$ at $(1, 0)$ and $(-1, 0)$
(on boundary)
min$= -1$ at $(0, 1), (0, -1)$
(on boundary)

9 Global min

11 Global max $= 0$
No global min

13 (a) Local min: $(-13/55, 1/55)$
(b)



15 All edges $(32)^{1/3}$ cm

17 $l = w = h = 45$ cm

19 $(3/14, 1/7, 1/14)$

23 $q_1 = 300, q_2 = 225.$

25 (a) $L = \left[pA\left(\frac{a}{k}\right)^a\left(\frac{l}{b}\right)^{(a-1)}\right]^{1/(1-a-b)}$
$K = \frac{la}{kb}L$
(b) No

27 $y = 24x^2/49 - 2/7$

31 (a) $p = \sqrt{P_0 P_F}$
(b) $p_1 = \sqrt[3]{P_0^2 P_F}, p_2 = \sqrt[3]{P_0 P_F^2}$

33 Must have global maximum if continuous and R closed, bounded

35 Local max not necessarily global max

37 $f(x, y) = x^2 + y^2$, R is $0 \le x, y \le 1$

39 False

41 False

43 False

45 True

Section 15.3

1 Min $= -\sqrt{2}$, max $= \sqrt{2}$

3 Max: 20 at $(-1, 2)$;
Min: 0 at $(1, -2)$

5 Min $= -22$, max $= 22$

7 Global min: $1/2$
No global max

9 Min $= -\sqrt{35}$, max $= \sqrt{35}$

11 Max: 4 at $(2, 2, 1)$,$(2, -2, -1)$, $(-2, 2, -1)$,$(-2, -2, 1)$;
Min: -4 at $(-2, -2, -1)$,$(-2, 2, 1)$, $(2, -2, 1)$,$(2, 2, -1)$

13 Max $= f(\frac{1}{\sqrt{5}}, \frac{3}{\sqrt{5}}) = 2\sqrt{5}$
Min $= f(-\frac{1}{\sqrt{5}}, -\frac{3}{\sqrt{5}}) = -2\sqrt{5}$

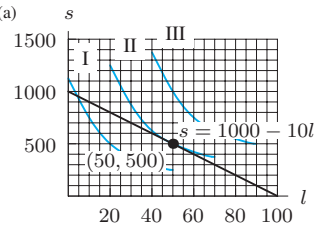
15 No global extrema

17 Max: $30 + \sqrt{10}$ at $(0, -\sqrt{10})$;
Min: -5 at $(\pm 3, -1)$

19 1 at $(1/2, 1/2)$

21 0.5

23 (a)



(b) $s = 1000 - 10l$

25 (a) Min; max at endpt of constraint
λ neg
(b) Max; min at endpt of constraint
λ pos

27 $\Delta c/4$; $-\Delta c/4$

31 (a) $C = \$4349$
(b) \$182

33 (a) $W = 225$
$K = 37.5$
(c) $W = 225$
$K = 37.5$
$\lambda = 0.29$

35 (a) No
(b) Yes
(c) $a + b = 1$

37 $x_1 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_1)^{1/2})$
$x_2 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_2)^{1/2})$

39 (a) $f_1 = \frac{k_1}{k_1+k_2}mg$, $f_2 = \frac{k_2}{k_1+k_2}mg$
(b) Distance the mass stretches the top spring and compresses the lower spring

43 (a) Cost of producing quantity u when prices are p, q
(b) $2\sqrt{pqu}$

45 (a) $-5\lambda^2 + 15\lambda$
(b) $1.5, 11.25$
(c) $11.25, 1.5$
(d) same

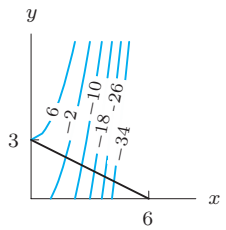
47 (a) $S = \ln(a^a(1-a)^{(1-a)}) + \ln b - a\ln p_1 - (1-a)\ln p_2$
(b) $b = e^c p_1^a p_2^{(1-a)}/(a^a(1-a)^{(1-a)})$

49 Maximum value is 1

51 $f(x, y) = 3x + 4y$

53 $f(x, y) = x^2 + y^2$

55



59 False

61 False

63 False

65 False

67 True

69 True

Chapter 15 Review

1 $(3, -1)$, Saddle point

3 Saddle pt: $(0, -5)$
Local min: $(2, -5)$

5 $(2, 1)$; local min

7 $(2, 3)$: Local and global max

9 Local minimum: $(1, 2)$
Local maximum: $(-1, -2)$
Saddle points: $(1, -2)$ and $(-1, 2)$
No global maximum or minimum

11 Minimum $f(-3/\sqrt{5}, 4/\sqrt{5}) = -5\sqrt{5}$,
Maximum $f(3/\sqrt{5}, -4/\sqrt{5}) = 5\sqrt{5}$

13 Min $= 11.25$; no max

15 Minimum $f(27.907, 23.256) = 1860.484$;
No maximum

17 Maximum $= f(48, 52) = 37{,}600$
No minimum

19 Max $= 21$, no min

21 No global extrema.

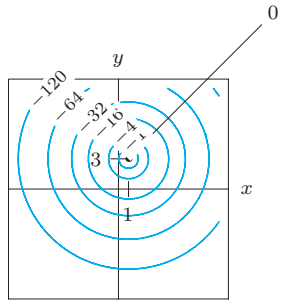
23 Maximum

25 Both

27 0.204

29 $\Delta c/2$; $\Delta c/(2\sqrt{2})$

31

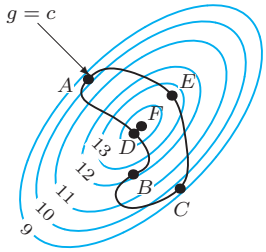


33 (a) (i) Power function
(ii) Linear function
(b) $\ln N = 1.20 + 0.32 \ln A$
Agrees with biological rule

35 (a) $2c^2/3$
(b) $4c/3$
(c) $\lambda = m'(c)$

37 $c\left(\frac{aK}{(a+b)P_1}\right)^a \left(\frac{bK}{(a+b)P_2}\right)^b$

39 (a) Points A, B, C, D, E
(b) Point F
(c) Point D



41 $f(x, y, z) = \frac{|Aa+Bb+Cc+D|}{\sqrt{A^2+B^2+C^2}}$

43 $K = 20$
$L = 30$
$C = \$7{,}000$

47 Base: 4 cm by 4 cm
Height: 2 cm

49 $-1/2$

51 (a) $x = 1/4 - a$, $y = 1$, saddle point

53 (a) 21.0208
(b) $g = 20.5, 21.2211$; $g = 20.2, 21.1008$
(c) $g = 20.5, 21.2198$; $g = 20.2, 21.1007$

Section 16.1

1 24; 43.5

3 Over: Approx 137
Under: Approx 60

5 about 2300

7 Positive

9 Zero

11 Zero

13 Positive

15 $25.2°\text{C}$

17 Need f nonnegative everywhere

19 $f(x, y) = 5 - x - y$; R is square with vertices $(\pm 1, \pm 1)$

21 False

23 False

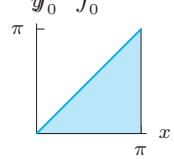
25 True

27 True

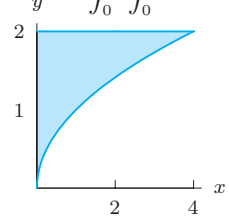
29 False

Section 16.2

1 $\int_0^{\pi}\int_0^{x} y \sin x \, dy \, dx$



3 $\int_0^{2}\int_0^{y^2} y^2 x \, dx \, dy$



5 150

7 54

9 $e - 2$

11 $3 - \sin 3$

13 $(e^4 - 1)(e^2 - 1)e$

15 -2.678

17 $\int_1^4\int_1^2 f \, dy \, dx$ or $\int_1^2\int_1^4 f \, dx \, dy$

19 $\int_{-1}^{3}\int_{-2}^{(1-3x)/4} f \, dy \, dx$
or $\int_{-2}^{1}\int_{-1}^{(1-4y)/3} f \, dx \, dy$

21 $\int_1^3 \int_{\frac{1}{2}(y-1)}^{-\frac{1}{2}(y-5)} f\,dx\,dy$

23 $\frac{4}{15}(9\sqrt{3} - 4\sqrt{2} - 1) = 2.38176$

25 $32/9$

27 $13/6$

29 0

31 $2/3$

33 $(e-1)/2$

35 $\frac{2}{9}(3\sqrt{3} - 2\sqrt{2})$

37 $\frac{1}{2}(e^2 - 1)$

39 (a) $8/3$
(b) $16/3$

41 (a) $\int_0^{1/2} \int_y^{1-y} f(x,y)\,dx\,dy$
$\int_0^{1/2} \int_0^{x} f(x,y)\,dy\,dx$
$\int_{1/2}^{1} \int_0^{1-x} f(x,y)\,dy\,dx$
(b) 1/8

43 15

45 $\int_{-5}^{5} \int_{-\sqrt{25-y^2}}^{\sqrt{25-y^2}} (25 - x^2 - y^2)\,dx\,dy$

47 $\int_0^4 \int_{y-4}^{(4-y)/2} (4 - 2x - y)\,dx\,dy$

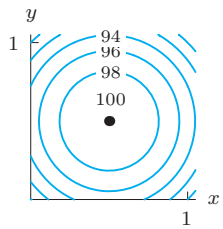
49 $1/3$

51 $16/3$

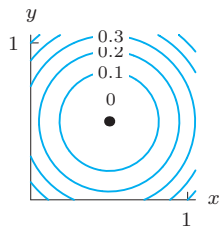
53 Volume $= 1/(6abc)$

55 (a) Circles centered at $(1, 0)$
(b) $\int_{-\sqrt{3}}^{\sqrt{3}} e^{-y^2}\,dy$
(c) $\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} e^{-(x-1)^2-y^2}\,dy\,dx$

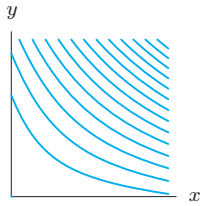
57 (a)



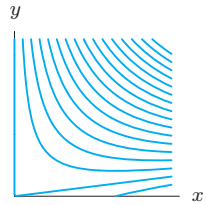
(b)



59 (a) $(4/3)a + b + (4/3)c = 20$
(b) $f(x,y) = x^2 + \frac{44}{3}xy + 3y^2$:



$f(x,y) = -3x^2 + 24xy$:



61 $k(a^3b + ab^3)/3$

63 Integrals not over same region

65 $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 2\,dx\,dy$

67 $\int_0^2 \int_0^{6-3y} 1\,dx\,dy$

69 False

71 False

73 False

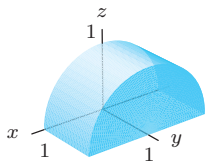
75 False

Section 16.3

1 2

3 -8

5



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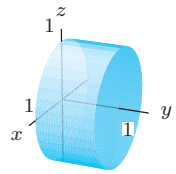

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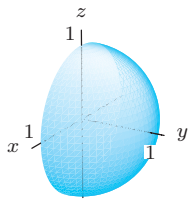

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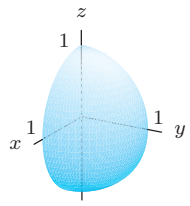

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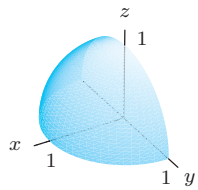


15 Positive

17 Positive

19 Zero

21 Positive

23 Zero

25 Positive

27 Positive

29 125

31 (a) $2\int_0^{10} \int_0^1 \int_y^1 e^{-3x}\,dz\,dy\,dx$
Other answers are possible
(b) $(1 - e^{-30})/3$

33 $V = \int_0^1 \int_0^2 \int_{x+y}^{1+2x+2y} 1\,dz\,dy\,dx$
Can reverse order x, y

35 $V = \int_0^1 \int_0^{1-x} \int_{6-3x-4y}^{6-2x-2y} 1\,dz\,dy\,dx$
Can reverse order x, y

37 $V = \int_{-\sqrt{5}}^{\sqrt{5}} \int_{-\sqrt{5-x^2}}^{\sqrt{5-x^2}} \int_2^{\sqrt{9-x^2-y^2}} 1\,dz\,dy\,dx$
Can reverse order x, y

39 $\int_0^1 \int_{-r}^{r} \int_0^{\sqrt{r^2-x^2}} f(x,y,z)\,dz\,dx\,dy$

41 $\int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \int_0^{\sqrt{r^2-x^2-z^2}} f(x,y,z)\,dy\,dz\,dx$

43 $1/2$

45 162

47 1

49 29 gm

51 243

53 (a) $x + y + z = 1$
(b) $1/6$

55 $\int_0^2 \int_0^{1-\frac{z}{4}} \int_0^{2-2y-\frac{z}{2}} f(x,y,z)\,dx\,dy\,dz$

57 $\int_{-4}^{4} \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} \int_3^{\sqrt{25-x^2-y^2}} dz\,dy\,dx$

59 $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} dy\,dz\,dx$

61 $\int_0^2 \int_0^{6y^2} \int_0^{\sqrt{12-3y^2}} f(x,y,z)\,dx\,dz\,dy$

63 4

65 $m = 2$;
$(\bar{x}, \bar{y}, \bar{z}) = (13/24, 13/24, 25/24)$

67 $5m/3$

71 Need same limits on innermost integral

73 $f(x,y,z) = z$

75 True

77 True

79 False

81 True

83 True

Section 16.4

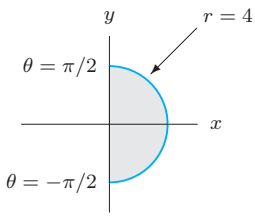
1 $\int_0^{\pi/2} \int_0^{1/2} f\,r\,dr\,d\theta$

3 $\int_{\pi/4}^{3\pi/4} \int_0^2 f\,r\,dr\,d\theta$

5 $\int_1^5 \int_2^4 f(x,y)\,dy\,dx$

7 $\int_\pi^{2\pi} \int_2^4 f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$

9



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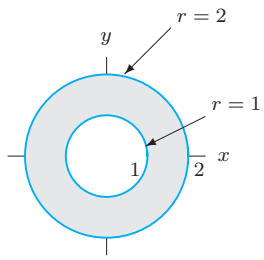

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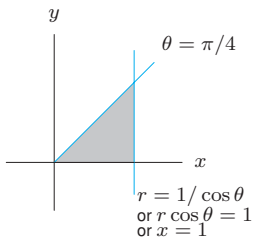

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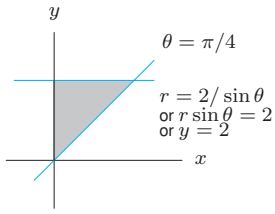


17 $\pi(1 - \cos 4)$

19 $-2/3$

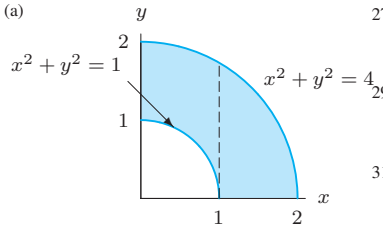
21 1.

23 (a) (i) $\int_0^{\pi/6} \int_0^{\pi} \int_0^5 \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$; $125\pi/9$

(ii) $\int_0^{\pi/6} \int_0^5 \int_{-\sqrt{25-r^2}}^{\sqrt{25-r^2}} r\,dz\,dr\,d\theta$; $2\int_0^{\pi/6} \int_0^5 \int_0^{\sqrt{25-z^2}} r\,dr\,dz\,d\theta$; $125\pi/9$

(b) $1/12(4/3\pi 5^3)$

25 (a)



(b) $7/3$

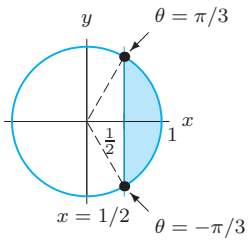
27 $32\pi(\sqrt{2} - 1)/3$

29 (a) $\int_0^{2\pi} \int_0^3 r/(r^2 + 1)\,dr\,d\theta$

(b) $\pi \ln 10$

31 $250\pi/3$ grams

33 (a)



(b) $(4\pi - 3\sqrt{3})/12$

35 (a) $\int_{-\sqrt{3}/2}^{\sqrt{3}/2} \int_{1-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dx\,dy$

(b) $\int_0^1 \int_{-\arccos(r/2)}^{\arccos(r/2)} r\,d\theta\,dr$

37 Upper limit for inner integral $1/\cos\theta$

39 Quarter disk $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$

41 (a), (c), (e)

Section 16.5

1 (a) is (IV); (b) is (II); (c) is (VII); (d) is (VI); (e) is (III); (f) is (V)

3 $z = \sqrt{1 - r^2}$

5 $\phi = \pi/4$

7 $\rho = 4/\cos\phi$

9 $200\pi/3$

11 25π

13 $\int_0^1 \int_0^{2\pi} \int_0^4 f \cdot r\,dr\,d\theta\,dz$

15 $\int_0^{\pi} \int_0^{\pi} \int_2^3 f \cdot \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

17 $\int_0^5 \int_0^2 \int_0^{x/5} f\,dz\,dy\,dx$

19 $\int_0^{2\pi} \int_0^K \int_{-\sqrt{K^2-r^2}}^{\sqrt{K^2-r^2}} r\,dz\,dr\,d\theta$

21 $\int_0^{2\pi} \int_0^2 \int_{2r}^4 f(r,\theta,z)\,r\,dz\,dr\,d\theta$

23 $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{2\sqrt{x^2+y^2}}^4 h(x,y,z)\,dz\,dy\,dx$

25 (a) $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{(1/2)-x^2}}^{\sqrt{(1/2)-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{1-x^2-y^2}} dz\,dy\,dx$

(b) $\int_0^{2\pi} \int_0^{1/\sqrt{2}} \int_r^{\sqrt{1-r^2}} r\,dz\,dr\,d\theta$

(c) $\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

27 (a) $\int_0^{2\pi} \int_0^{\sqrt{2}} \int_r^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$

(b) $\int_0^{2\pi} \int_0^{\pi/4} \int_0^2 \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

29 $V = \int_0^{2\pi} \int_0^{\pi/3} \int_0^3 \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$
Order of integration can be altered; other coordinates can be used

31 $V = \int_0^{\pi} \int_{\sqrt{2}}^{\sqrt{3}} \int_5^{10} r\,dz\,dr\,d\theta$;
Order of integration can be altered; other coordinates can be used

33 $V = \int_0^{2\pi} \int_0^3 \int_1^{\sqrt{10-r^2}} r\,dz\,dr\,d\theta$
or
$V = \int_0^{2\pi} \int_1^{\sqrt{10}} \int_0^{\sqrt{10-z^2}} r\,dr\,dz\,d\theta$ Order of integration can be altered; other coordinates can be used

35 (a) $\int_0^{2\pi} \int_0^{1/\sqrt{3}} \int_{\sqrt{3}r}^1 r\,dz\,dr\,d\theta$

(b) $\pi/9$

37 $16\pi(\sqrt{2} - 1)/(3\sqrt{2})$

39 $28\pi/15$

41 $\int_0^{2\pi} \int_0^{5/\sqrt{2}} \int_r^{5/\sqrt{2}} r\,dz\,dr\,d\theta = 125\pi/(6\sqrt{2}) = 46.28\ \text{cm}^3$

43 (a) Negative

(b) Zero

45 $pqr/6$

47 $\int_0^{2\pi} \int_0^{\pi/2} \int_a^b \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta = 2\pi(b^3 - a^3)/3$

49 (a) $\int_0^{2\pi} \int_0^{\sqrt{2}} \int_0^{2-r^2} (2 - z)r\,dz\,dr\,d\theta$

(b) $\left(\frac{16\sqrt{2}}{3} - \frac{4}{3}\right)\pi$

51 $\int_0^{2\pi} \int_0^{\pi} \int_0^3 2\rho^3 \sin\phi\,d\rho\,d\phi\,d\theta$

53 $1/32$

55 (a) Mass $= \int_0^{2\pi} \int_0^a \int_{-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} k|z|r\,dz\,dr\,d\theta$

(b) $\pi k a^4/2$

57 $3/4$

61 $3I = \frac{6}{5}a^2$; $I = \frac{2}{5}a^2$

63 $(q^2/8\pi\epsilon)((1/a) - (1/b))$

65 Mass $= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{4-x^2-y^2} e^{-x-y}\,dz\,dy\,dx$ gm

67 $1/27$

69 $2\pi G\delta(H + R - \sqrt{R^2 + H^2})$

71 Total charge $= 2\pi k R^2$

73 (c)

75 Limits of outer integral not constant

77 $\int_W \sqrt{x^2 + y^2 + z^2}\,dx\,dy\,dz$;
W is unit ball $x^2 + y^2 + z^2 \le 1$

Section 16.6

1 Is a joint density function

3 Not a joint density function

5 Is joint density function

7 0

9 1

11 7/8

13 1/16

15 (a) 20/27

(b) 199/243

17 (a) $k = 3/8$

(b) 15/32

(c) 1/16

19 $\int_{65}^{100} \int_{0.8}^1 f(x,y)\,dx\,dy$

23 $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (p_1(x,y) + p_2(x,y))\,dx\,dy = 2$

25 $a = 0,\ b = 1,\ c = 0,\ d = 1$

27 True

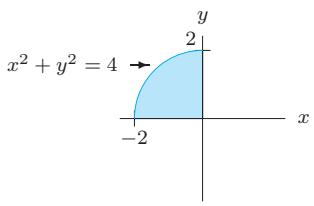
29 False

Chapter 16 Review

1 94.5

3 14

5



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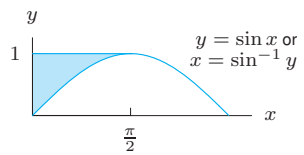
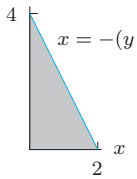


11 $\int_0^{2\pi} \int_0^2 \int_0^3 f r\, dr dz d\theta$

13 $\int_0^{\pi/2} \int_0^{\pi} \int_0^5 f \rho^2 \sin\phi\, d\rho d\phi d\theta$

15 (a) y



(b) $\int_0^2 \int_0^{-2x+4} g(x, y)\, dy\, dx$

17 $10(e - 2)$

19 $(1/20)\sin^5 1$

21 $(\pi/2)(1 - e^{-1})$

23 1/48

25 $\int_0^{2\pi} \int_{3\pi/4}^{\pi} \int_0^{\sqrt{2}} f(\rho, \phi, \theta)\rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$

27 $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{2-x^2-y^2}}^{-\sqrt{x^2+y^2}} h(x, y, z)\, dz\, dy\, dx$

29 Positive

31 Positive

33 Negative

35 Zero

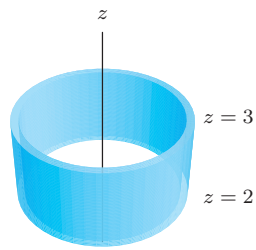
37 Positive

39 1.571

41 (a) $\int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_0^{3/\sin\theta} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$

(b) 18π

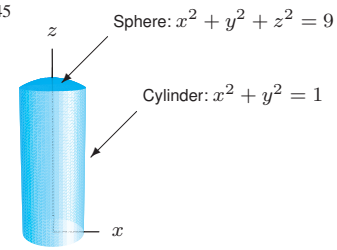
43



$\int_0^{2\pi} \int_2^3 \int_{\sqrt{5}}^{\sqrt{6}} r\, dr\, dz\, d\theta$

Order of integration can be changed

45



$\int_0^{2\pi} \int_0^1 \int_0^{\sqrt{9-r^2}} r\, dz\, dr\, d\theta$

47 Negative

49 Can't tell

51 Zero

53 Zero

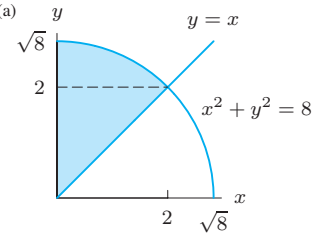
55 Zero

57 Negative

59 $\pi(3 - 2\ln 2)$

61 $162\pi/5$

63 (a)



(b) $\pi(1 - e^{-8})/8$

65 (a) $z = \sqrt{1 - (y-1)^2}$ $0 \le y \le 2$ $0 \le x \le 10$

(b) $\int_0^{10} \int_0^2 \int_0^{\sqrt{1-(y-1)^2}} f(x, y, z)\, dz\, dy\, dx$

67 (a) Half cylinder, radius 1, along x-axis, $y \ge 0, -1 \le x \le 1$

(b) $2\pi/5$

69 $\int_0^{2\pi} \int_0^4 \int_3^{\sqrt{25-r^2}} r\, dz\, dr\, d\theta$
$= 52\pi/3 = 54.45\ \text{cm}^3$

71 $\int_0^{2\pi} \int_0^{\arccos(3/5)} \int_{3/\cos\phi}^{5} \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$
$= 52\pi/3 = 54.45\ \text{cm}^3$

73 8π gm

75 $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-1+\sqrt{2-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz dy dx$
$= 2\pi((7/6) - 2\sqrt{2}/3)$

77 $8\pi R^5/15$

79 (a) $3/2000$
(b) $343/1000$

81 $6\pi/7$

83 $a, b + c, 0$

Section 17.1

1 $x = 0, y = t, -2 \le t \le 1$

3 $x = t, \quad y = 3 - 3t, \quad 0 \le t \le 1$

5 $x = 1 + 2t, \quad y = 1 + t, \quad 0 \le t \le 1$

7 $x = t, y = 1, z = -t$

9 $x = 1, \quad y = 0, \quad z = t$

11 $x = 1 + 3t, \quad y = 2 - 3t, \quad z = 3 + t$

13 $x = -3 + 2t, y = -2 - t, z = 1 - 2t$

15 $x = 2 + 3t, \quad y = 3 - t, \quad z = -1 + t$

17 $x = 3 - 3t, y = 0, z = -5t$

19 $x = 3\cos t, y = 3\sin t, z = 5, 0 \le t < 2\pi$

21 $x = 2\cos t, y = -2\sin t, z = 0$

23 $x = 2\cos t, y = 0, z = 2\sin t$

25 $x = 0, y = 3\cos t, z = 2 + 3\sin t$

27 $x = t^2, y = t, z = 0$

29 $x = -3t^2, y = 0, z = t$

31 $x = t, y = 4 - 5t^4, z = 4$

33 $x = 3\cos t, y = 2\sin t, z = 0$

35 $x = -1 + 3t, y = 2, z = -3 + 5t$

37 $\vec{r}(t) = \vec{i} - 3\vec{j} + 2\vec{k} + t(3\vec{i} + 4\vec{j} - 5\vec{k})$, $0 \le t \le 1$, $x = 1 + 3t$, $y = -3 + 4t$, $z = 2 - 5t$, $0 \le t \le 1$

39 $x = \cos t, y = \sin t, z = 0, 0 \le t \le \pi$

41 Two arcs:
$\vec{r}(t) = 5\vec{i} + 5(-\cos t\vec{i} + \sin t\vec{j})$, $0 \le t \le \pi$ or
$\vec{r}(t) = 5\vec{i} + 5(\cos t\vec{i} + \sin t\vec{j})$, $\pi \le t \le 2\pi$

43 $\vec{r}(t) = (2 + 10t)\vec{i} + (5 + 4t)\vec{j}$

45 $\vec{r}(t) = (2 + ((t - 20)/10)10)\vec{i}$
$+(5 + ((t - 20)/10)4)\vec{j}$

47 $\vec{r}(t) = (2 - 10t)\vec{i} + (5 - 4t)\vec{j}$

49 No

51 (b) $-\vec{i} - 10\vec{j} - 7\vec{k}$
(c) $\vec{r} = (1 - t)\vec{i} + (3 - 10t)\vec{j} - 7t\vec{k}$

53 (a) $\vec{r} = (\vec{i} + 3\vec{j} + 7\vec{k}) + t(2\vec{i} - 3\vec{j} - \vec{k})$
(b) $(3, 0, 6)$
(c) $\sqrt{14}$

55 (a) $2\vec{i} - 5\vec{j} + 3\vec{k}$
(c) $x = 1 + 2t, y = -1 - 5t, z = 1 + 3t$

57 Different lines

59 Different lines

61 (a) $(-1, 4, -2)$
(b) $-\vec{i} - \vec{j} + 2\vec{k}$; other answers possible
(c) $\vec{r} = -\vec{i} + 4\vec{j} - 2\vec{k} + t(-\vec{i} - \vec{j} + 2\vec{k})$

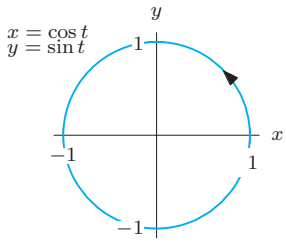
63 $x = -\frac{3}{2}t + \frac{5}{2}, y = \frac{1}{2}t + \frac{1}{2}, z = t.$

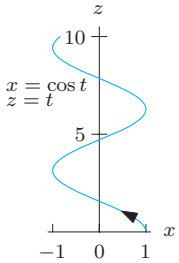
65 $x = -4, y = 2 + t, z = 3 + t$

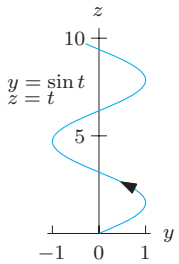
67 (a) Straight lines
(b) No
(c) $(1, 2, 3)$

69 (a) Repeats every year
(b) Mid-August
(c) Mid-April
(d) 2°C per month

71 Circle, cosine, sine







73 (a) $x = 2 + at, y = 1 + bt, z = 3 + ct$
(b) $a/2 = b/1 = c/3$

75 α is radius of cylinder
β determines vertical distance moved

77 (a) (vii)
(b) (ii)
(c) (iv)

79 $\vec{r} = 9.65\vec{k} + t(325\vec{i} + 563\vec{j} - 0.84\vec{k})$

81 (a) Parallel
(b) (i) Perpendicular
(ii) Parallel

83 Different parameterization of same curve

85 $x = \cos t, y = \sin t, z = 0$
$x = 0, y = \cos t, z = \sin t$

87 $x = t^3, y = 2t^3, z = 3 + 4t^3$

89 True

91 True

93 True

95 True

97 False

99 False

Section 17.2

1 $\vec{v} = 3\vec{i} + \vec{j} - \vec{k}, \quad \vec{a} = \vec{0}$

3 $\vec{v} = \vec{i} + 2t\vec{j} + 3t^2\vec{k}, \quad \vec{a} = 2\vec{j} + 6t\vec{k}$

5 $\vec{v} = -3\sin t\vec{i} + 4\cos t\vec{j}$,
$\vec{a} = -3\cos t\vec{i} - 4\sin t\vec{j}$

7 $\vec{v} = \vec{i} + 2t\vec{j} + 3t^2\vec{k}$,
Speed $= \sqrt{1 + 4t^2 + 9t^4}$,
Particle never stops

9 $\vec{v} = 6t\vec{i} + 3t^2\vec{j}$,
$\|\vec{v}\| = 3|t| \cdot \sqrt{4 + t^2}$,
Stops when $t = 0$

11 $\vec{v} = 6t\cos(t^2)\vec{i} - 6t\sin(t^2)\vec{j}$,
$\|\vec{v}\| = 6|t|$,
Stops when $t = 0$

13 Length $= \sqrt{42}$

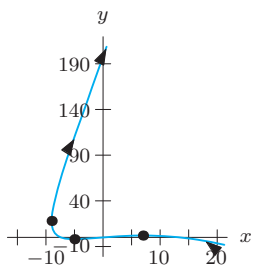
15 Length $= e - 1$

17 $\vec{v} = -6\pi\sin(2\pi t)\vec{i} + 6\pi\cos(2\pi t)\vec{j}$,
$\vec{a} = -12\pi^2\cos(2\pi t)\vec{i} - 12\pi^2\sin(2\pi t)\vec{j}$,
$\vec{v} \cdot \vec{a} = 0, \|\vec{v}\| = 6\pi, \|\vec{a}\| = 12\pi^2$

19 Line through $(2, 3, 5)$ in direction of
$\vec{i} - 2\vec{j} - \vec{k}$,
$\vec{v} = 2t(\vec{i} - 2\vec{j} - \vec{k}), \vec{a} = 2(\vec{i} - 2\vec{j} - \vec{k})$

21 $x = 1 + 2(t - 2), y = 2, z = 4 + 12(t - 2)$

23 Vertical: $t = 3$
Horizontal: $t = \pm 1$
As $t \to \infty, x \to \infty, y \to \infty$
As $t \to -\infty, x \to \infty, y \to -\infty$



25 (a) $\vec{v}(2) \approx -4\vec{i} + 5\vec{j}$,
Speed $\approx \sqrt{41}$
(b) About $t = 1.5$
(c) About $t = 3$

27 (a) $x = 2 + 0.6t, y = -1 + 0.8t, z = 5 - 1.2t, 0 \le t \le 5$
(b) $x = 2 + 1.92t, y = -1 + 2.56t, z = 5 - 3.84t, 0 \le t \le 1.56$

29 (a) 6.4 meters
(b) 1.14 sec
(c) 15.81 m/sec
(d) $(11.4, -5.7, 0)$
(e) -9.8 m/sec^2

31 (a) 5 secs; $(10, 15, 100)$
(b) $t = 0$, 10 secs, $\sqrt{113}$ cm/sec
(c) 5 secs, $\sqrt{13}$ cm/sec

33 (a) $t = 5.181$ sec
(b) $x = 103.616$ meters
(c) 2 meters
(d) 9.8 meters/sec^2
(e) $\theta = 0.896$; $v = 32.016$ meters/sec

35 (a) (IV); 4.5 sec; $(0, 8.9$ m$, 0)$
(b) (II); 3.2 sec; base of tower
(c) (V); 10 sec; halfway up

37 $\vec{r}(t) = 22.1t\vec{i} + 66.4t\vec{j}$
$+ (442.7t - 4.9t^2)\vec{k}$

41 (a) No
(b) $t = 5$
(c) $\vec{v}(5) \approx 0.959\vec{i} + 0.284\vec{j} + 2\vec{k}$
(d) $\vec{r} \approx 0.284\vec{i} - 0.959\vec{j} + 10\vec{k}$
$+ (t - 5)(0.959\vec{i} + 0.284\vec{j} + 2\vec{k})$.

43 (a) $\vec{r}(t) = t\cos(2\pi t)\vec{i} + t\sin(2\pi t)\vec{j}$,
$0 \le t \le 100$
(b) $\vec{v} = \vec{r}\,'(t) =$
$(\cos(2\pi t) - 2\pi t\sin(2\pi t))\vec{i} +$
$(\sin(2\pi t) + 2\pi t\cos(2\pi t))\vec{j}$
$\|\vec{v}\| = (1 + 4\pi^2 t^2)^{1/2}$ cm/sec
(c) $\vec{a} = \vec{v}\,'(t) =$
$(-4\pi\sin(2\pi t) - 4\pi^2 t\cos(2\pi t))\vec{i} +$
$(4\pi\cos(2\pi t) - 4\pi^2 t\sin(2\pi t))\vec{j}$
$\|\vec{a}\| = 4\pi(1 + \pi^2 t^2)^{1/2}$ cm/sec^2

45 (a) $x \approx 694.444t$
$+ 20\cos(2\pi t)$cm
$y = 30 + 20\sin(2\pi t)$ cm
(c) At least 5.526 revs/sec

47 No

49 (a) $2\vec{r} \cdot d\vec{r}/dt$
(b) $\vec{a} \times d\vec{r}/dt$
(c) $r^3 d\vec{r}/dt + 3r^2\vec{r}$

55 Acceleration is a vector, not a scalar

57 $\vec{r}(t) = \left(t + 2t^2\right)\vec{i} + 2t\vec{j} + 3t^2\vec{k}$

59 False

61 False

63 False

65 False

67 True

69 False

Section 17.3

1 $\vec{V} = x\vec{i}$

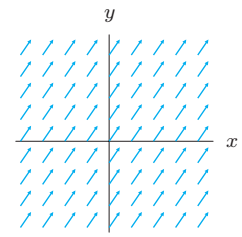
3 $\vec{V} = x\vec{i} + y\vec{j} = \vec{r}$

5 $\vec{V} = -x\vec{i} - y\vec{j} = -\vec{r}$

7 (a) y-axis
(b) Increasing
(c) Neither

9 (a) x-axis
(b) Increases
(c) Decreases

11



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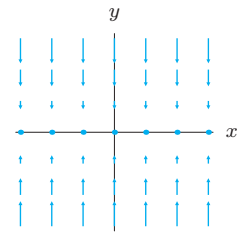

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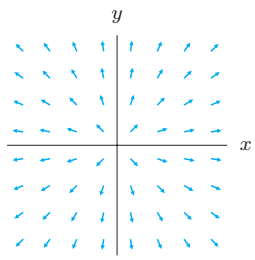

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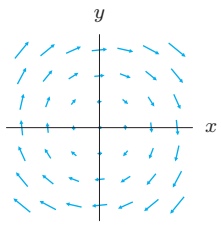

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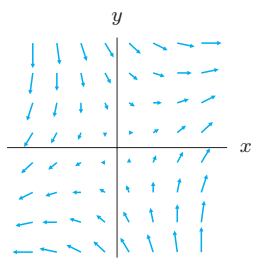


21 (a) IV
(b) III
(c) I
(d) II

23 $(1+x^2)(3\vec{i}+2\vec{j})$, other answers possible

25 $x\vec{i}+y\vec{j}$, other answers possible

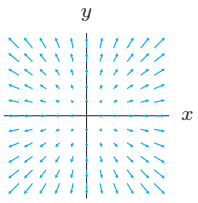
27 (A) = (IV); (B) = (II); (C) = (I); (D) = (III)

29 (a) II
(b) III
(c) IV
(d) I

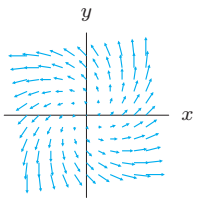
31 $\vec{F}(x,y)=\frac{-x\vec{i}-y\vec{j}}{\sqrt{x^2+y^2}}$ (for example)

33 (a) $(1,-3,-7)$; other answers possible
(b) $(0,0,0)$; other answers possible
(c) $-4x+y-3z=0$; plane through origin

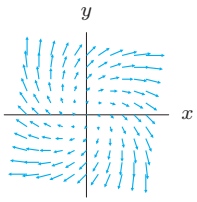
35 (a) Radiates out from origin



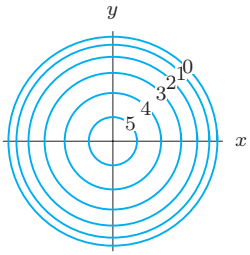
(b) Spirals outward counterclockwise around origin



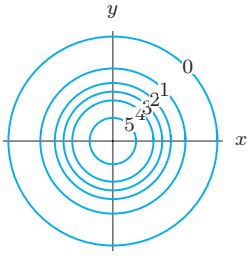
(c) Spirals outward clockwise around origin



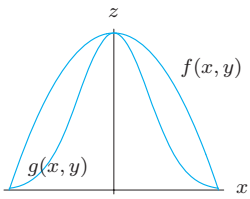
37 (a) $z=f(x,y)$:



$z=g(x,y)$:



(b)

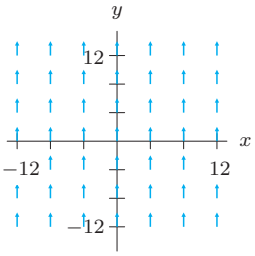


39 To plot $\vec{G}(x,y,z)$ move arrows of $\vec{F}(2x,2y,2z)$ halfway to origin

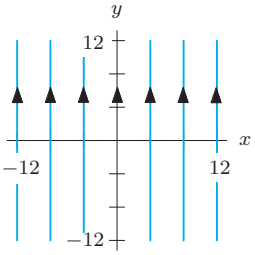
41 $(x^2+1)\left(\vec{i}+\vec{j}+\vec{k}\right)$

Section 17.4

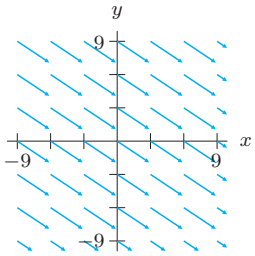
1 Field:



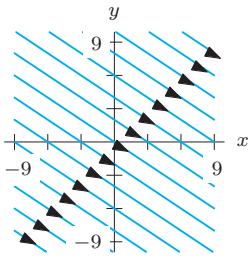
Flow, $x=$ constant:



3 Field:



Flow, $y=-(2/3)x+c$:



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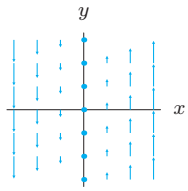
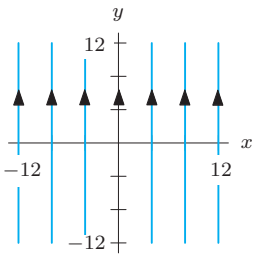
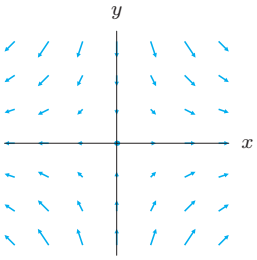


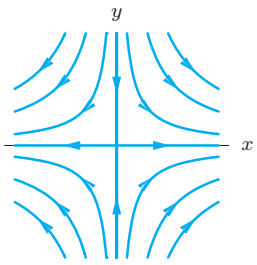
Flow:



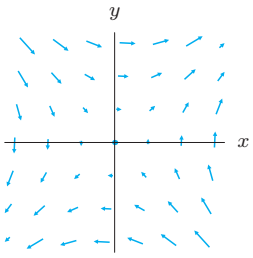
7 Field:



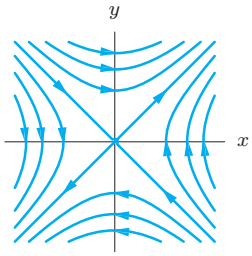
Flow:



9 Field:



Flow:



15 (a) Same directions, different magnitudes
(b) Same curves, different parameterizations

23 (a) $h'(t) = 0$

25 Counterexample: $\vec{F}(x, y, z) = (x^2 + 1)\vec{i}$

27 $\vec{F}(x, y, z) = x\vec{i} + y\vec{j} + z\vec{k}$

29 False

31 False.

33 False

35 False

37 False

Chapter 17 Review

1 $\vec{r} = 2\vec{i} - \vec{j} + 3\vec{k} + t(5\vec{i} + 4\vec{j} - \vec{k})$

3 $x = t,\ y = 5$

5 $x = 4 + 4\sin t, y = 4 - 4\cos t$

7 $x = 2 - t, y = -1 + 3t, z = 4 + t.$

9 $x = 1 + 2t, y = 1 - 3t, z = 1 + 5t.$

11 $x = 3\cos t$
$y = 5$
$z = -3\sin t$

13 $\vec{r} = 10\cos(2\pi t/30)\vec{i} - 10\sin(2\pi t/30)\vec{j} + 7\vec{k}$

15 $\vec{v} = \vec{i} + (3t^2 - 1)\vec{j}$

17 $\vec{v} = 6t\vec{i} + 2t\vec{j} - 2t\vec{k}$

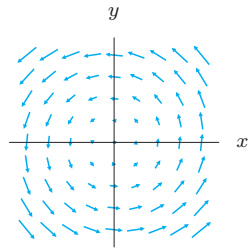
19 Vector; $\left((3\cos\sqrt{2t+1})\vec{i} - (3\sin\sqrt{2t+1})\vec{j} + \vec{k}\right)/\sqrt{2t+1}$

21 Vector; $-(\cos t/(2\sqrt{3+\sin t}))\vec{i} - (\sin t/(2\sqrt{3+\cos t}))\vec{j}$

23 No

25 Same direction $-\vec{i} + 4\vec{j} - 2\vec{k}$, point $(3, 3, -1)$ in common

27



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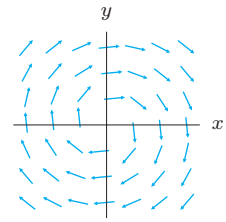


31 (a) $(2, 3, 0)$
(b) 2
(c) No; not on line

33 (b) (i) Yes
(ii) No
(iii) Yes
(iv) No

35 $\vec{E}$: (IV); $\vec{F}$: (I);
$\vec{G}$: (II); $\vec{H}$: (III)

37 $\vec{v} = 0.2\vec{i} - 0.4\vec{j} + 0.4\vec{k}$ m/sec
$\vec{a} = \vec{0}$

39 (a) Max rate change temp with distance; $^\circ$C/cm
(b) $\sqrt{(g'(t))^2 + (k'(t))^2}$; cm/sec
(c) $f_x \cdot g'(t) + f_y \cdot k'(t)$; $^\circ$C/sec

41 (a) Yes, $t = 1, (x, y) = (-2, -1)$
(b) Yes, $t = -1, (x, y) = (2, 3)$
(c) No

43 (a) $x(t) = 5\sin t, y(t) = 5\cos t, z(t) = 8$
(b) $\vec{v} = -5\vec{j}, \vec{a} = -5\vec{i}$
(c) $x_{tt}(t) = y_{tt}(t) = 0,$
$z_{tt}(t) = -g,$
$x_t(0) = z_t(0) = 0,$
$y_t(0) = -5,\ x_t(0) = 5,\ y_t(0) = 0,$
$z_t(0) = 8$

45 (a) In the direction given by the vector: $\vec{i} - \vec{j}$
(b) Directions given by unit vectors:
$\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$
$-\frac{1}{\sqrt{2}}\vec{i} - \frac{1}{\sqrt{2}}\vec{j}$
(c) -4

47 $-(2 + 4t^2)/(t^3(1 + t^2)^2)$

49 (a) $52/\sqrt{13}$, $(8, 12)$
(b) $52/\sqrt{13}$, $(18, 62)$

51 No, since the point $(0, 1)$ is not on the curve

53 ω: Rate of change of polar angle θ of particle,
a: Rate of change of particle's distance from origin

55 (a) $\frac{1}{x^2+y^2+z^2}$
(b) $\frac{1}{\sqrt{x^2+y^2+z^2}}$
(c) $\frac{x}{\sqrt{x^2+y^2+z^2}}\vec{i} + \frac{y}{\sqrt{x^2+y^2+z^2}}\vec{j} + \frac{z}{\sqrt{x^2+y^y+z^2}}\vec{k}$
(d) $\frac{-x}{\sqrt{x^+y^2+z^2}}\vec{i} + \frac{-y}{\sqrt{x^2+y^2+z^2}}\vec{j} + \frac{-z}{\sqrt{x^2+y^y+z^2}}\vec{k}$
(e) $\frac{\cos t}{2\sqrt{2}}\vec{i} + \frac{\sin t}{2\sqrt{2}}\vec{j} + \frac{1}{2\sqrt{2}}\vec{k}$
(f) $\frac{1}{\sqrt{2}}$

57 No

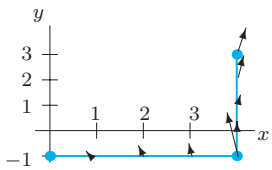
59 (b) $\vec{i} + 2\vec{j} + 3\vec{k} + 5\cos t((1/\sqrt{2})\vec{i} - (1/\sqrt{2})\vec{j}) + 5\sin t((1/\sqrt{6})\vec{i} + (1/\sqrt{6})\vec{j} - (\sqrt{2}/\sqrt{3})\vec{k})$

61 (b) $e^{-t}\vec{i} - 2e^{-t}\vec{j}$,
$(0.0025e^{3t} + 0.9975e^{-t})\vec{i}$
$+(0.005e^{3t} - 1.995e^{-t})\vec{j}$,

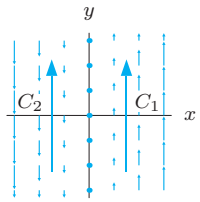
$(-0.0025e^{3t} + 1.0025e^{-t})\vec{i}$
$+(-0.005e^{3t} + 2.005e^{-t})\vec{j}$

Section 18.1

1 Negative
3 Zero
5 Zero
7 0
9 0
11 28
13 16
15 -48
17 $19/3$
19 20
21 28
23 C_1 is pos; C_2, C_3 are zero
25 C_1 is neg; C_2 is neg; C_3 is zero
27 $\int_{C_3} \vec{F} \cdot d\vec{r} < \int_{C_1} \vec{F} \cdot d\vec{r} < \int_{C_2} \vec{F} \cdot d\vec{r}$
29 (b)



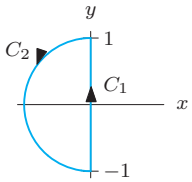
(c) 60
31 $a > 0$
33 $b < 0$
35 $c > 3$
37 0
39 Negative
41 0
43 Yes
49 $-GMm/8000$
51 Sphere of radius a centered at the origin
53 (a) $\phi(\vec{r}) = -\dfrac{Q}{4\pi\epsilon}\dfrac{1}{a} + \dfrac{Q}{4\pi\epsilon}\dfrac{1}{||\vec{r}||}$
(b) Because then $\phi(\vec{r}) = \dfrac{Q}{4\pi\epsilon}\dfrac{1}{||\vec{r}||}$
55 Value of a line integral is not a vector
57



59 True
61 True
63 False
65 False
67 False

Section 18.2

1 $\int_0^{\pi} (\cos^2 t - \sin^2 t)\, dt$
Other answers are possible
3 $\int_0^{2\pi} (-\sin t \cos(\cos t) + \cos t \cos(\sin t))dt$
5 24
7 -4
9 -6
11 9
13 $82/3$
15 12
17 116.28
19 12
21 21
23 0
25 $\int_C y^2 dx + z^2 dy + (x^2 - 5)dz$
27 $e^{-3y}\vec{i} - yz(\sin x)\vec{j} + (y + z)\vec{k}$
29 $18\pi^2$
31 -18π
33 (a)



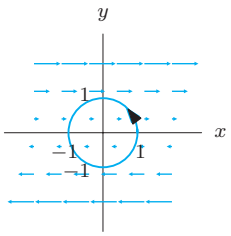
(b) $0; -3\pi/2$
37 (a) -5
(b) 5
(c) 0
41 (a) Greater than zero
(b) 88
(d) $t = 2 \pm 1/\sqrt{3}$
(e) 88; yes
43 Sign depends on C
45 $y = \pi/2, x = t, 0 \le t \le 3, \int_C \vec{F} \cdot d\vec{r} = 3$
47 True
49 True
51 False
53 False

Section 18.3

1 12
3 Negative, not path-independent
5 Negative, not path-independent
7 Path-independent
9 Path-independent
11 Path-independent
13 $f(x, y) = x^2 y + K$
15 $f(x, y, z) = e^{xyz} + \sin(xz^2) + C$
$C =$ constant
17 -2
19 2
21 0
23 $e^3 - 1$
25 0
27 PQ
31 Yes
33 Yes.
35 $e^9 - 1$
37 (a) $-(1/3) - \pi$
(b) $3\pi/4$
39 $9\pi/4$
41 $e^{9/2} + \sin(3\sqrt{2}) - 1$
43 107; 109.5
45 -2
47 (b) No
49 (a) $\int_{C_2} \vec{F} \cdot d\vec{r}$
(b) $\int_{C_3} \vec{F} \cdot d\vec{r} < \int_{C_2} \vec{F} \cdot d\vec{r}$
$< \int_{C_4} \vec{F} \cdot d\vec{r} < \int_{C_1} \vec{F} \cdot d\vec{r}$
(c) $\int_{C_3} \vec{F} \cdot d\vec{r} = -\int_{C_4} \vec{F} \cdot d\vec{r} < 0$
51 -3.6
53 (a) $\vec{a}$
(b) $\vec{r}_0 \cdot \vec{a}$
(c) $10||\vec{a}||$
55 (a) $\pi/2$
(b) No
57 (a) $\vec{F} - \text{grad}\, \phi = x\, \text{grad}\, h$
(b) 50
59 (a) $\vec{F} - \text{grad}\, \phi = x\, \text{grad}\, h$
(b) 384
63 (b) Yes
65 Only if $\vec{F}$ is path-independent
67 $\vec{F} = \text{grad}\, f, f(x, y) = 50xy$
73 False
75 False
77 False
79 True
81 True
83 False

Section 18.4

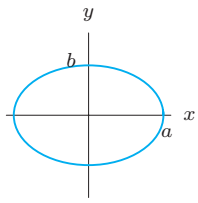
1 No
3 No
5 $f(x, y) = x^3/3 + xy^2 + C$
7 Yes, $f = \ln A|xyz|$ where $A > 0$
9 No
11 -12
13 $1/2$
15 -3π
17 (a)



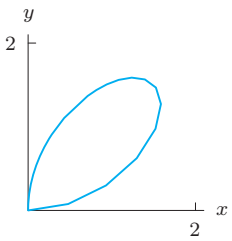
(b) $-\pi$
19 $e - \cos 1$
21 $-9\pi/8$
23 (a) 0
(b) 0
(c) 0
(d) -6π
(e) -6π
(f) 0
(g) -6π

25 (b) 0
(c) $\vec{G} = \nabla(xyz + zy + z)$
(d) $\vec{H}_1 = \nabla(yx^2), \vec{H}_2 = \nabla(y(x+z))$

27 πab



29 $3/2$



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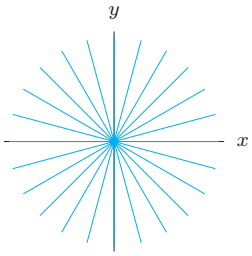


33 (a) Possible answers are:
$\vec{F} = \text{grad}(xy)$
$\vec{G} = \text{grad}(\arctan(x/y)),\ y \neq 0$
$\vec{H} = \text{grad}\left((x^2+y^2)^{1/2}\right),\ (x,y) \neq (0,0)$
(b) $0, -2\pi, 0$
(c) Does not apply to $\vec{G}$, $\vec{H}$; holes in domain

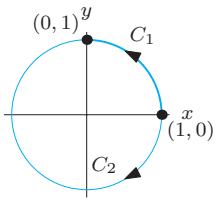
35 $L_1 < L_2 < L_3$

37 (a) $21\pi/2$
(b) 2

39 (a) $\vec{0}$
(b) $q/||\vec{r}\,||$

41 Green's Theorem does not apply;
Line integral depends on $\vec{F}$

43



45 True

47 True

49 True

51 True

Chapter 18 Review

1 Positive

3 (a) Zero
(b) C_1, C_3: Zero
C_2: Negative
C_4: Positive
(c) Zero

5 Scalar; 12

7 -58

9 50

11 18

13 1372

15 Not path-independent

17 Path-independent

19 Path-independent

21 Path-independent

23 350

25 $27\pi/2$

27 45

29 0, 100

31 (ii), (iv)

33 (a) 24
(b) 12
(c) -12

37 18

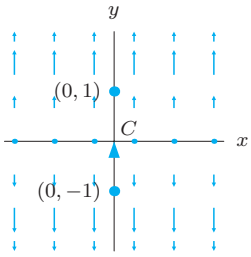
39 -36

41 (a) 0
(b) 24

43 (a) 0
(b) 6
(c) $75\pi/2$
(d) 14

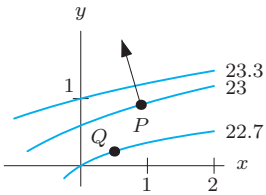
45 (a) $9/2$
(b) $-9/2$

47



49 (a) Closed curve oriented counterclockwise
(b) Closed curve oriented clockwise with $y > 0$ or
Closed curve oriented counterclockwise if $y < 0$
(Other answers are possible)

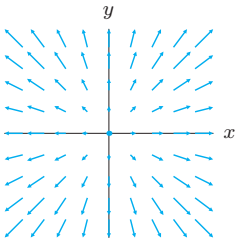
51 (a)



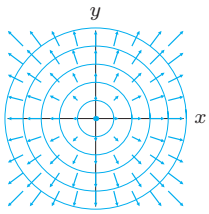
(b) Longer
(c) -0.3

53 (a) $\pi/2$
(b) 0

57 (a) $\omega = 3000$ rad/hr
$K = 3 \cdot 10^7\ \text{m}^2\cdot\text{rad/hr}$
(b) Inside tornado:



View from great distance:



(c) $r < 100$ m, circulation is $2\omega\pi r^2$
$r \geq 100$ m, circulation is $2K\pi$

59 (a) $-(\pi/2)(-6a^2 + a^4)$; $a = \sqrt{3}$
(b) Integrand $3 - x^2 - y^2$ positive inside disk of radius $\sqrt{3}$, negative outside

61 $18a + 18b + 36c + (81d/2)$,
$-18a - 18b - 36c - (81d/2)$,
curves go in opposite directions

Section 19.1

1 $-3\vec{\imath}$

3 $15\vec{\jmath}$

5 Rectangle in xz plane with area 150, oriented pos y direction

7 (a) 45
(b) -45

9 (a) Positive
(b) Negative
(c) Zero
(d) Zero
(e) Zero

11 (a) Zero
(b) Zero
(c) Zero
(d) Negative
(e) Zero

13 8

15 -6

17 4

19 $3/2$

21 2π

23 0

25 3

27 108π

29 $9\pi\sqrt{75}$

31 2π

33 28

35 28π

37 0

39 114

41 $64/3$

43 756π

45 18

47 $81\pi/2$

49 $\pi(e^4 - 1)$

51 8π

53 (a) (i) Positive
(ii) Zero
(iii) Zero
(iv) Negative
(b) Integral over B

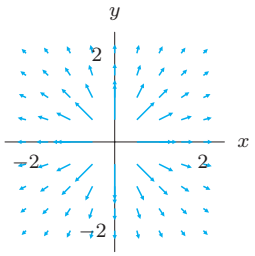
55 $6\pi\ \text{cm}^3/\text{sec}$

57 $\vec{j}$ components parallel to S, other components same

59 $4\pi R$

61 (a) Zero
(b) Zero

65 (a)



(b) $4\pi\lambda h$

67 (a) 0
(b) 0

69 Sign of $\int_S \vec{F} \cdot d\vec{A}$ depends on both $\vec{F}$ and S

71 $\vec{F} = z\vec{k}$
S: $0 \le x \le 1, 0 \le y \le 1, z = 1$, oriented upwards

73 False

75 True

77 True

79 False

81 False

Section 19.2

1 $\left(-3\vec{i} + 5\vec{j} + \vec{k}\right) dx\, dy$

3 $\left(-4x\vec{i} + 6y\vec{j} + \vec{k}\right) dx\, dy$

5 $\int_{-2}^{3}\int_{0}^{5} 70\, dy\, dx$

7 $\int_0^5\int_0^{5-x}(yz\sin x - 2xy\cos 2y + xy)\, dy\, dx$

9 -500

11 $-5/3 - \sin 1 = -2.508$

13 $\int_0^{\pi/2}\int_0^5 10\,(\cos\theta + 2\sin\theta)\, dz\, d\theta$

15 $\int_0^{2\pi}\int_{-8}^{8}\left(6z^2\cos\theta + 6\sin\theta e^{6\cos\theta}\right) dz\, d\theta$

17 2000

19 $100\sqrt{2}/3$

21 $\int_0^{2\pi}\int_0^{\pi/2} 100\,(\sin\phi\cos\theta + 2\sin\phi\sin\theta + 3\cos\phi)\sin\phi\, d\phi\, d\theta$

23 $\int_{-\pi/2}^{\pi/2}\int_0^{\pi} 16\cos^2\phi\sin^2\phi\cos\theta\, d\phi\, d\theta$

25 $8000/3$

27 $(8 - 5\sqrt{2})\pi/6 = 0.486$

29 6

31 6

33 18

35 36π

37 $7/3$

39 $\pi\sin 25$

41 $\pi/2$

43 1296π

45 $1/4$

47 36π

49 $2\pi/3$

51 $4\pi a^3$

53 -1

55 $11\pi/2$

57 (a) Constant inside sphere of radius a

(b) $\vec{E} = \begin{cases} k\dfrac{\delta_0}{3}\rho\vec{e}_\rho & \rho \le a \\ k\dfrac{\delta_0 a^3}{3r^3}\vec{e}_\rho & \rho > a \end{cases}$

59 Cone not on a cylinder

61 $f(x, y) = -x - y$

63 True

65 False

Section 19.3

1 $2x + xe^z$

3 (I)

5 0

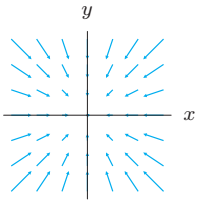
7 -1

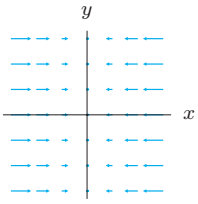
9 $6x - (x + z)\cos(xz)$

11 0

13 $2/\|\vec{r} - \vec{r}_0\|$

15





17 -0.030

19 (a) (i) $0.016\pi/3$
(ii) -0.08
(b) Flux positive at $(2, 0, 0)$ and negative at $(0, 0, 10)$

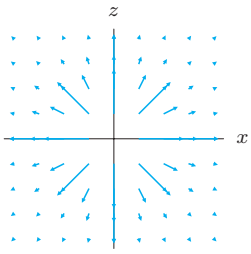
21 (a) 4
(b) 4
(c) 4

23 $\operatorname{div}\vec{v} = -6$

25 (a) $-1/3, 1$
(b) $1/3$

27 (b)

29 (a) 0
(b)

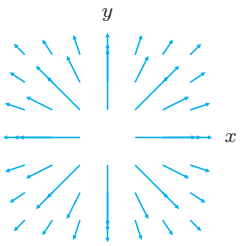


31 (a) 0
(b) Undefined

33 (a) $\rho(0) < \rho(1000) < \rho(5000)$
(b) cars/hour
(d) $\rho(x) = 4125/(55 - x/50)$
if $0 \le x < 2000$
$\rho(x) = 4125/15 = 275$
if $2000 \le x < 7000$
$\rho(x) = \dfrac{4125}{(15 + (x - 7000)/25)}$
if $7000 \le x < 8000$
$\rho(x) = 4125/55 = 75$
if $x \ge 8000$
(e) 139 ft. at $x = 0$
89 ft. at $x = 1000$
38 ft. at $x = 5000$

39 $(3 - a)x^{-a}$

43 (d)



45 $\operatorname{div}(2x\vec{i}) = 2$

47 Only vector fields have divergence

49 $\vec{F}(x, y, z) = y^2\vec{i} + xz\vec{j} + x\vec{k}$

51 True

53 True

55 True

57 False

59 False

61 True

63 False

Section 19.4

1 24

3 8

5 Zero

7 24

9 72

11 288

13 36π

15 620π

17 48π

19 1.571

21 Yes; -3.22

23 $\int_S \vec{F} \cdot d\vec{A} = \int_W \operatorname{div} \vec{F}\, dV = 0$

25 (a) 0
(b) No
(c) 4π
(d) 4π

27 (a) $cb(12a - a^2)$
(b) 6, 10, 10; 3600

29 4π

31 (a) $24\pi/5$
(b) $\int_{S_2} \vec{F} \cdot d\vec{A} < \int_{S_3} \vec{F} \cdot d\vec{A} < \int_{S_4} \vec{F} \cdot d\vec{A}$

33 S closed, oriented inward

39 $\int_S \vec{F} \cdot d\vec{A} = \int_W \operatorname{div} \vec{F}\, dV$

41 $\vec{F} = (9x/4\pi)\vec{i}$

43 True.

45 False.

47 False.

49 True

51 True

53 True

55 True

Chapter 19 Review

1 Scalar; -100π

3 (a) 16
(b) 16
(c) -16
(d) -16
(e) 32

5 Negative

7 Positive

9 0

11 -45

13 27/2

15 -4

17 12

19 $-8(1 + e^{-1})$

21 24π

23 $(\pi/6) - 1/3$

25 $c > 0$

27 $a + b + c > 0$

29 (a) 24
(b) 24

31 1500π

33 $8\pi/3$

35 2π

37 (a) $(y + yze^{xyz})\vec{i} + (x + xze^{xyz})\vec{j} + xye^{xyz}\vec{k}$
(b) $5 + e^{24} - e^1$
(c) 2

39 (a) Flux through S_3
(b) 13π

41 (a) (i) 0.0168
(ii) -0.004
(i) Pointing outward
(ii) Pointing inward

43 (a) Flux $= c^3$
(b) 1
(c) 1

45 -10π

47 0

49 34π

51 (a) 0
(b) z-axis
(c) 0
(d) 0

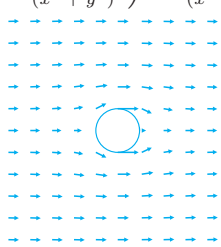
53 (a) 0
(b) 0

55 (a) $x = 2$, $\vec{i}$ direction
(b) (i) 24π
(ii) 8π

57 81π

61 (a) Maximum speed
(b) 0
(c) $\pi u a^2/2$

63 (a) $\left(1 + \dfrac{y^2 - x^2}{(x^2 + y^2)^2}\right)\vec{i} + \dfrac{-2xy}{(x^2 + y^2)^2}\vec{j}$
(d)



65 (a) 0.0958
(b) 0.0958
(c) 0.0958

Section 20.1

1 Vector: $\vec{i} + \vec{j} - \vec{k}$

3 $6\vec{i}$

5 $-\vec{i} - \vec{j} - \vec{k}$

7 $\vec{0}$

9 $(2x^3yz + 6x^7y^5 - xy)\vec{i} + (-3x^2y^2z - 7x^6y^6 + y)\vec{j} + (yz - z)\vec{k}$

11 Nonzero curl

13 Zero curl

15 (a) $-2y\vec{i} + 2x\vec{j}$
(b) $4\vec{k}$

17 (a) 0.0005π
(b) 0.0002π
(c) -0.0003π

19 $\operatorname{curl}(F_1(x)\vec{i} + F_2(y)\vec{j} + F_3(z)\vec{k}) = \vec{0}$

21 (a) Counterclockwise
(b) Clockwise
(c) $\vec{0}$

33 (b) Direction of $\vec{k}$ for $A = -1$
Zero vector for $A = -2$
Direction of $-\vec{k}$ for $A = -3$
(c) At $(1, 1, 1)$: Positive for $A = -1$
Zero for $A = -2$
Negative for $A = -3$
At $(0, 0, 0)$: Can't say anything

35 Counterexample: $\vec{F} = y\vec{i}$

37 $\vec{F} = z\vec{i}$

39 True.

41 True

43 False

45 False

Section 20.2

1 $9\pi/2$

3 25

5 $-\pi$

7 -5π

9 6

11 $-6\sqrt{3}\pi$

13 Negative

15 (a) $-\vec{i} - \vec{j} - \vec{k}$
(b) -6

17 (a) $\vec{i} + \vec{j} + \vec{k}$
(b) $-4\sqrt{3}$

19 $8\pi/\sqrt{3}$

21 0

23 $8\sqrt{3}\pi$

25 (a) $x^2 + y^2 = 9, z = 2$
(b) -18π

27 72

31 120π

33 (a) Parallel to xy-plane; same in all horizontal planes
(b) $(\partial F_2/\partial x - \partial F_1/\partial y)\vec{k}$
(d) Green's Theorem

35 C not the boundary of a surface

37 Any oriented circle

39 True

41 False

43 True

45 True

47 False

Section 20.3

1 Yes

3 Yes

5 No

7 Yes

9 Yes

11 Yes

13 Curl yes; Divergence yes

15 Curl yes; Divergence yes

19 $(1/2)\vec{b} \times \vec{r}$

21 No

23 (a) Yes
(b) Yes
(c) Yes

25 (a) Yes
(b) No
(c) No

29 (b) $\nabla^2\psi = -\operatorname{div} \vec{A}$

31 Curl of scalar function not defined

33 $f(x, y, z) = x^2$

35 False

37 True

Chapter 20 Review

1 $\vec{i} - \vec{j} - \vec{k}$

3 (c), (d), (f)

5 C_2, C_3, C_4, C_6

7 Defined; scalar

9 Defined; vector

11 Nonzero curl

13 div $\vec{F} = y + z + x$
curl $\vec{F} = -y\vec{i} - z\vec{j} - x\vec{k}$
$\vec{F}$ not solenoidal; not irrotational

15 div $\vec{F} = 0$
curl $\vec{F} = (2y - \cos(x+z))\vec{i} - \left(2x - e^{y+z}\right)\vec{j} + \left(\cos(x+z) - e^{y+z}\right)\vec{k}$
$\vec{F}$ is solenoidal; not irrotational

17 (a) 18π
(b) 18π

19 20π

21 0

23 (a) is (I); (b) is (I); (c) is (V); (d) is (III); (e) is (IV); (f) is (V)

25 (a) -1125π
(b) Not defined
(c) 0
(d) Not defined
(e) $384\pi/5$
(f) $353/4$
(g) Not defined
(h) 54

27 (a) 23
(b) Approx $0.0003\pi/\sqrt{3}$

29 (a) div $\vec{F} < 0$, div $\vec{G} < 0$
(b) curl $\vec{F} = 0$, curl $\vec{G} = 0$
(c) Yes
(d) Yes
(e) No
(f) No

31 0

33 4π

35 210

37 $12\pi/5$

39 150π

41 $e - 1$

43 (a) 18
(b) $81/2$

45 (a) No
(b) No

47 (a) 2π
(b) $\vec{0}$ except on z-axis
(c) No
(d) Yes; 0
(e) No

49 (a) 50
(b) 30π
(c) 8
(d) -18π

51 (a) $(4(p+q)\pi R^3/3)$
(b) $(4(p+q)\pi R^3/3)$

Section 21.1

1 Curve

3 Surface

5 Horizontal disk of radius 5 in plane $z = 7$

7 Helix radius 5 about z-axis

9 Top hemisphere

11 Vertical segment

13 $\vec{r}(s,t) = (s+2t)\vec{i} + (2s+t)\vec{j} + 3s\vec{k}$, other answers possible

15 $\vec{r}(s,t) = (3+s+t)\vec{i} + (5-s)\vec{j} + (7-t)\vec{k}$, other answers possible

17 (a) Yes
(b) No

19 $s = s_0$: lines parallel to y-axis with $z = 1$
$t = t_0$: lines parallel to x-axis with $z = 1$

21 $s = s_0$: parabolas in planes parallel to yz-plane
$t = t_0$: parabolas in planes parallel to xz-plane

23 $s = 4, t = 2$
$(x, y, z) = (x_0 + 10, y_0 - 4, z_0 + 18)$

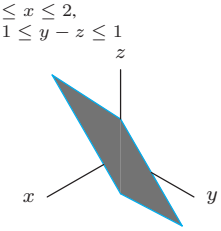
25 Horizontal circle

27 (a) $x = \left(\cos\left(\frac{\pi}{3}t\right) + 3\right)\cos\theta$
$y = \left(\cos\left(\frac{\pi}{3}t\right) + 3\right)\sin\theta$
$z = t \quad 0 \le \theta \le 2\pi,\ 0 \le t \le 48$
(b) 456π in.^{3}

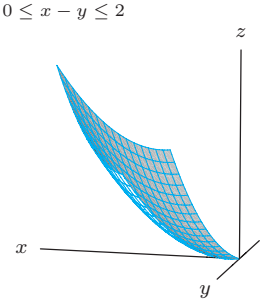
29 If $\theta < \pi$, then $(\theta + \pi, \pi/4)$
If $\theta \ge \pi$, then $(\theta - \pi, \pi/4)$

31 $x = r\cos\theta, \quad 0 \le r \le a$
$y = r\sin\theta, \quad 0 \le \theta \le 2\pi$
$z = (1 - r/a)\,h$

33 (a) $-x + y + z = 1$,
$0 \le x \le 2$,
$-1 \le y - z \le 1$
(b)



35 (a) $z = (x^2/2) + (y^2/2)$
$0 \le x + y \le 2$
$0 \le x - y \le 2$
(b)



37 Radius: $R\sin\phi$

39 $x + y - z - 3 = 0$

41 True

43 True

45 True

47 False

Section 21.2

1 1

3 e^{2s}

5 $a = 1/10, b = 1$

7 $a = 1/50, b = 1/10$

9 3

11 $\rho^2 \sin\phi$

13 13.5

15 72

17 (a) $(1/(2\pi\sigma^2)\int_{-\infty}^{\infty}\int_{-\infty}^{2t-x} e^{-(x^2+y^2)/(2\sigma^2)}\,dy\,dx$
(b) $(1/(\sqrt{\pi}\sigma))\int_{-\infty}^{t} e^{-u^2/\sigma^2}\,du$
(c) $(1/(\sqrt{\pi}\sigma))e^{-t^2/\sigma^2}$
(d) Normal, mean 0, standard deviation $\sigma/\sqrt{2}$

19 R does not correspond to T

21 $x = 2s, y = 3t$

23 False

Section 21.3

1 $((s+t)\vec{i} - (s-t)\vec{j} - 2\vec{k})\,ds\,dt$

3 $-e^s(\cos t\,\vec{j} + \sin t\,\vec{k})\,ds\,dt$

5 4/3

7 $6(e^4 - 1)$

9 $-\pi R^7/28$

11 $200\sqrt{14}$

13 $\sqrt{6}\pi$

15 $khw^3/6$ meter^3/sec.

21 Integral gives volume

23 $\vec{r}(s,t) = 2s\vec{i} + t\vec{j}$

25 True

27 False

Chapter 21 Review

1 Cone, height 7 and radius 14

3 $a = 1/15, b = 1/15$

5 $-1/2$

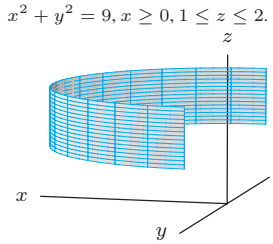
7 0

9 $x = 2 + 5\sin\phi\cos\theta$
$y = -1 + 5\sin\phi\sin\theta$
$z = 3 + 5\cos\phi$

11 (a) Cylinder
(b) Helices

13 $x = a\sin\phi\cos\theta \quad 0 \le \phi \le \pi$
$y = b\sin\phi\sin\theta \quad 0 \le \theta \le 2\pi$
$z = c\cos\phi$

15 (a) $x^2 + y^2 = 9, x \ge 0, 1 \le z \le 2.$
(b)



17 0

19 $2\pi c(a^2 + b^2)$

Appendix A

1 (a) $y \le 30$
(b) two zeros

3 -1.05

5 2.5

7 $x = -1.1$